THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

Royal School of Engineering and Technology (RSET)

Department of Computer Science and Engineering (CSE)

Course Structure & Syllabus

(Based on National Education Policy 2020)

For

Bachelor of Technology

In

Artificial Intelligence & Data Science

W.E.F

AY: 2026-2027

Section 6

Course Structure and Syllabus of the Framework

6.1 Course Structure of B. Tech (AI & DS)

1st Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Basic Science Course (BSC)							
1	PHY022C101	Physics	3	0	0	3	3
2	PHY022C121	Physics Lab	0	0	2	1	2
3	MAT022C102	Mathematics – I	3	1	0	4	4
Engineering Science Course (ESC)							
4	CSE022C103	Basic Electrical Engineering	3	0	0	3	3
5	CSE022C113	Basic Electrical Engineering Lab	0	0	2	1	2
6	CEE022C114	Engineering Graphics & Design Lab	0	0	4	2	4
7	CSE022C105	Programming for Problem Solving -I	2	0	0	2	2
8	CSE022C115	Programming for Problem Solving -I Lab	0	0	2	1	2
Humanities/Social Science including Management Course (HSMC)							
9	BHS022A101	Universal Human Values: Understanding Harmony and Ethical Human Conduct	3	0	0	3	3
Mandatory Courses (MC)/ Skill Enhancement Courses							
10	CSE022S116	Design Thinking and Ideation Lab	0	0	2	1	2
		TOTAL	**14**	**1**	**12**	**21**	**27**
2nd Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Basic Science Course (BSC)							
1	CHY022C201	Chemistry	3	0	0	3	3
2	CHY022C211	Chemistry Lab	0	0	2	1	2
3	MAT022C202	Mathematics – II	3	1	0	4	4
4	CSE022C203	Basic Environmental Engineering	3	0	0	3	3
Engineering Science Course (ESC)							
5	CSE022C204	Programming for Problem Solving	3	0	0	3	3
6	CSE022C214	Programming for Problem Solving Lab	0	0	2	1	2
7	MEE022C215	Manufacturing Practices Workshop	0	0	4	2	4
Humanities/Social Science including Management Course (HSMC)							
8	CEN022A201	English for Technical Writing	2	0	0	2	2

Mandatory Courses (MC)/ Skill Enhancement Courses							
9	CSE022S217	Sports and Yoga	0	0	2	1	2
		TOTAL	**14**	**1**	**10**	**20**	**25**
3rd Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	MAT022C301	Discrete Mathematics	3	1	0	4	4
2	CSE022C302	Data Structures and Algorithms	3	0	0	3	3
3	CSE022C312	Data Structures and Algorithms Lab	0	0	2	1	2
4	CSE022C303	Computer Organisation and Architecture	3	0	0	3	3
Engineering Science Course (ESC)							
5	CSE022C304	Digital Logic and Design	3	0	0	3	3
6	CSE022C314	Digital Logic and Design Lab	0	0	2	1	2
IKS							
7	IKS992K301	Indian Knowledge System-I	2	0	0	2	2
Open Elective							
8	CSE022C305	OOP using C++	3	0	0	3	3
9	CSE022C315	OOP using C++ Lab	0	0	2	1	2
Internship							
10	CSE022C327	Internship-I	0	0	0	2	0
		TOTAL	**17**	**1**	**6**	**23**	**23**
4th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C401	JAVA Programming	3	0	0	3	3
2	CSE022C411	JAVA Programming Lab	0	0	2	1	2
3	CSE022C402	Database Management Systems	3	0	0	3	3
4	CSE022C412	Database Management Systems Lab	0	0	2	1	2
5	CSE022C403	Formal Language and Automata Theory	3	1	0	4	4
6	ARI022C406	Mathematical Foundations for AI	4	0	0	4	4
Open Elective							
7	CSE022C405	Programming with Python	3	0	0	3	3
8	CSE022C415	Programming with Python Lab	0	0	2	1	1
Internship							
9	CSE022C427	Internship-II	0	0	0	1	0
		TOTAL	**16**	**1**	**4**	**20**	**21**
5th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C501	Operating Systems	3	0	0	3	3
2	CSE022C511	Operating Systems Lab	0	0	2	1	2
3	ARI022C502	Introduction to Artificial Intelligence	3	0	0	3	3
4	ARI022C502	Introduction to Artificial Intelligence Lab	0	0	2	1	2
5	CSE022C503	Design and Analysis of Algorithms	3	0	0	3	3

6	CSE022C513	Design and Analysis of Algorithms Lab	0	0	2	1	2
7	CSE022C505	Data Communication	3	0	0	3	3
Humanities/Social Science including Management Course (HSMC)							
8	BSA022C505	Principles of Management & Organisational Behaviour	3	0	0	3	3
Open Elective							
9	CSE022G506	Open Elective-III (Prompt Engineering)	3	0	0	3	3
Internship							
10	CSE022C527	Internship-III	0	0	0	2	0
		TOTAL	**18**	**0**	**6**	**23**	**25**
6th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C601	Computer Networks	3	0	0	3	3
2	CSE022C611	Computer Networks Lab	0	0	2	1	2
3	ARI022C602	Introduction to ML and DL	3	0	0	3	3
4	ARI022C612	Introduction to ML and DL Lab	0	0	2	1	2
5	CSE022C603	Software Engineering	3	0	0	3	3
Professional Elective Course (PEC)							
6	ARI022D60X	Professional Elective Course-I	4	0	0	4	4
7	ARI022D60X	Professional Elective Course-II	4	0	0	4	4
Open Elective							
8	CSE022G606	Open Elective-IV (Entrepreneurship and Startup)	3	0	0	3	3
Internship							
9	CSE022C627	Internship-IV	0	0	0	1	0
		TOTAL	**20**	**0**	**4**	**23**	**24**
7th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	ARI022C703	Data and Visual Analytics in AI	3	0	0	3	3
2	ARI022C713	Data and Visual Analytics in AI Lab	0	0	2	1	2
3	CSE022C702	Web Technology/ Applied AI	3	0	0	3	3
4	CSE022C712	Web Technology Lab/ Applied AI Lab	0	0	2	1	2
Professional Elective Course (PEC)							
5	ARI022D70X	Professional Elective Course-III	4	0	0	4	4
6	ARI022D70X	Professional Elective Course-IV	4	0	0	4	4
Internship							
7	CSE022C725	Internship-V	0	0	0	4	0
Project/ Dissertation							
8	CSE022C726	Project-I	0	0	4	2	4
		TOTAL	**13**	**0**	**2**	**22**	**22**
8th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	ARI022C801	Predictive Modelling & Optimization Techniques	3	0	0	3	3

2	ARI022C811	Predictive Modelling & Optimization Techniques Lab	0	0	2	1	2
3	CSE022C802	Cryptography and Network Security	3	0	0	3	3
4	CSE022C812	Cryptography and Network Security Lab	0	0	2	1	2
Professional Elective Course (PEC)							
4	ARI022D80X	PEC-V	4	0	0	4	4
Project/ Dissertation							
5	CSE022C824	Project-II	0	0	12	6	12
		TOTAL	**11**	**0**	**14**	**18**	**26**

PEC Tracks	Subjects
Artificial Intelligence	PEC 1: Advanced Deep Learning (ARI022D601)
	PEC 2: Natural Language Processing (ARI022D602)
	PEC 3: Computer Vision (ARI022D701)
	PEC 4: Reinforcement Learning (ARI022D702)
	PEC 5: AI for Robotics (ARI022D801)
Data Analytics	PEC 1: Statistical Thinking for Data Science (ARI022D603)
	PEC 2: Big Data Analytics (ARI022D604)
	PEC 3: Pattern Recognition & Visual Recognition (ARI022D703)
	PEC 4: Bioinformatics (ARI022D704)
	PEC 5: AI for Gaming (ARI022D802)

List of Open Electives to be offered by Department of CSE	
Open Elective-I	Programming with Python (CSE022G306)
Open Elective-II	Fundamentals of Web Design (CSE022G406)
Open Elective-III	Introduction to AI (CSE022G506)/ Prompt Engineering
Open Elective-IV	Fundamentals of IOT (CSE022G606)/ Entrepreneurship and Startups

6.2 Detailed Syllabus of 1st Semester

Paper I/Subject Name: Physics **Subject Code: PHY022C101**

L-T-P-C – 3-0-0-3 **Credit Units: 03** **Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students enhance the fundamental knowledge in Physics and its applications relevant to various streams of Engineering and Technology

Prerequisites: Concepts of Physics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic terminologies of physics	**BT 1**
CO 2	**Understand** the basic concepts of Physics	**BT 2**
CO 3	**Solve** problems in wave mechanics.	**BT 3**
CO 4	**Analyze** knowledge in calculating red and blue shift and also in acoustics.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I.	**Electromagnetic Theory**	Electrostatics in vacuum, Electrostatics in a linear dielectric medium, Magneto statics, Magneto statics in a linear magnetic medium, Faraday's law, Displacement current, Magnetic field due to time-dependent electric field and Maxwell's equations, Electromagnetic waves	**09**
II.	**Introduction to Mechanics**	Transformation of scalars and vectors under Rotation transformation; Potential energy function; Non-inertial frames of reference; Harmonic oscillator; Definition and motion of a rigid body in the plane; Rotation in the plane; Kinematics in a coordinate system rotating and translating in the plane; Angular momentum, Introduction to three-dimensional rigid body motion	**12**
III.	**Quantum Mechanics for Engineers**	Wave nature of particles and the Schrodinger equation, Mathematical Preliminaries for quantum mechanics, Applying the Schrodinger equation, molecular bonding, Solids	**06**
IV	**Oscillations, Waves and Optics**	Simple harmonic motion, damped and forced simple harmonic oscillator, Non-dispersive transverse and longitudinal waves in one dimension and introduction to dispersion, The propagation of light and geometric optics, Wave optics, Lasers	**09**
		TOTAL	**36**

Credit Distribution

Lecture/ Tutorial	Practicum	Experiential Learning
3 * 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Elements of properties of matter,* Mathur D.S., 7th Edition, Revised Edition, 2005, S. Chand publication, New Delhi.
2. *Electricity and Magnetism,* Tayal D.C, Publisher, 4th Edition, 2017, Himalaya Publishing House,New Delhi.
3. *Geometrical and Physical Optics,* Chakraborty P.K., 3rd Edition, 2005, New Central Book agency (P) Ltd.

Reference Books:

1. Singh A.K. and Malik Hitendra *Engineering Physics,* 2nd Edition, 2016, McGraw Hill Education Private Limited. New Delhi.
2. Gaur R.K and Gupta S.L, *Engineering Physics*, 2015, Dhanpat Rai publication, New Delhi.
3. Arthur Beiser, Shobhit Mahajan, S. Rai. Choudhury, *Concept of Modern physics,* 6th Edition, 2009, McGraw-Hill education Private limited. New Delhi.
4. M Ghosh & D Bhattacharya, *A Textbook of Oscillations, Waves and Acoustics,* 5th Edition, 2016, S. Chand publication.

Additional Readings

1. https://www.griet.ac.in/nodes/Engineering%20Physics%20Notes.pdf
2. https://mrcet.com/downloads/digital_notes/HS/R20/Engineering%20Physics.pdf
3. NPTEL Course on Introduction to Electromagnetic Theory by Prof. Manoj Harbola, IIT Kanpur

Paper II/Subject Name: Physics Lab		**Subject Code: PHY022C111**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students familiarize with fundamental principles of physics through experiments relevant to computing and electronics, develop basic experimental, data recording, and analytical skills in measurement and observation, etc.

Prerequisites: Concepts of Physics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** Identify and explain the working principles of optical, electrical, and electronic instruments.	**BT 1 & 2**
CO 2	**Perform** experiments accurately using standard laboratory equipment and follow scientific procedures.	**BT 3**
CO 3	**Analyze** experimental data, calculate errors, and interpret results effectively.	**BT 4**

CO 4	**Compare** theoretical predictions with experimental results and discuss deviations.	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Experiment	Experiment Title	Lab Hours
I	Determination of Moment of Inertia of a given solid about its own axis by using M.I.Table	2
II	Determination of Young's Modulus using Searle's Apparatus	2
III	Determination of Rigidity of Modulus of the material of the given rod by Statiscal method	2
IV	Determination of Powers of Given lenses using an Optical Bench i. Concave Lens, ii Convex Lens	2
V	Determination of Resistance of a Galvanometer using Post Office Box.	2
VI	To determine the mechanical equivalent of heat by Joule's calorimeter	2
VII	Determination of ratio of E.M.F of two cells using Potentiometer.	2
VIII	To determination of the focal length of a convex mirror with the help of an auxiliary lens.	2
IX	Determination of Horizontal Components of Earth's Magnetic field using Magnetometer	2
X	Determination of coefficient of Viscosity of water by Capillary Flow Method	2

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Elements of properties of matter,* Mathur D.S., 7th Edition, Revised Edition, 2005, S. Chand publication, New Delhi.
2. *Electricity and Magnetism,* Tayal D.C, Publisher, 4th Edition, 2017, Himalaya Publishing House,New Delhi.
3. *Geometrical and Physical Optics,* Chakraborty P.K., 3rd Edition, 2005, New Central Book agency (P) Ltd.

Reference Books:

1. Singh A.K. and Malik Hitendra *Engineering Physics,* 2nd Edition, 2016, McGraw Hill Education Private Limited. New Delhi.
2. Gaur R.K and Gupta S.L, *Engineering Physics*, 2015, Dhanpat Rai publication, New Delhi.
3. Arthur Beiser, Shobhit Mahajan, S. Rai. Choudhury, *Concept of Modern physics,* 6th Edition, 2009, McGraw-Hill education Private limited. New Delhi.
4. M Ghosh & D Bhattacharya, *A Textbook of Oscillations, Waves and Acoustics,* 5th Edition, 2016, S. Chand publication.

Paper III/Subject Name: Mathematics-I		Subject Code: MAT022C102
L-T-P-C – 3-1-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to enable students to achieve conceptual understanding and to retain the best traditions of traditional calculus

Prerequisites: Concepts of Mathematics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** basic terminologies of calculus	**BT 1**
CO 2	**Understand** the applications of differential and integral calculus in different fields of Engineering.	**BT 2**
CO 3	**Apply** the single and multivariable differential and Integral calculus in engineering problems.	**BT 3**
CO 4	**Analyze** and **assess** the patterns in series	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Basic & Single Variable Calculus**	Curvature, evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. Rolle's Theorem, Mean value theorems and applications; Extreme values of functions; Linear approximation; Indeterminate forms and L' Hospital's rule.	**12**
II.	**Multi Variable Calculus I**	Limit, continuity and partial derivatives, directional derivatives, gradient, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers.	**12**
III.	**Multi Variable Calculus II**	Multiple Integration: Double integrals (Cartesian), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian), orthogonal curvilinear coordinates, Simple applications involving cubes, sphere and rectangular parallelepipeds; Scalar line integrals, vector line integrals, scalar surface integrals, vector surface integrals, Gradient, curl and divergence, Theorems of Green, Gauss and Stokes.	**12**
IV	**Sequence and Series**	Limits of sequence of numbers, Calculation of limits, Infinite series; Tests for convergence; Power series, Taylor and Maclaurin series; Taylor theorem, convergence of Taylor series, error estimates.	**12**

TOTAL	**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4*12 NCH = 48 NCH	--	72 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *A text book of Engineering Mathematics*, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
2. *Mathematical Methods for Physics and Engineering: A Comprehensive Guide,* K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press

Reference Books:

1. Grewal B. S., *Higher Engineering Mathematics*, 43rd Edition, 2014, Khanna Publishers.
2. Das B. C. & Mukherjee B. N., *Differential Calculus*, 55th Edition, U. N. Dhur & Sons Pvt. Ltd.
3. Das B. C. & Mukherjee B. N, *Integral Calculus*, 57th Edition, U. N. Dhur & Sons Pvt. Ltd

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Mathematics-I.
2. https://www.vidyalankar.org/gate/assets/docs/notes/maths.pdf

Paper IV/Subject Basic Electrical Engineering		**Subject Code: CSE022C103**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make students understand the basic electrical terminologies and familiarize them with the basic concepts of D.C., single-phase and three-phase A.C. networks.

Prerequisites: Basic concepts of D.C. networks of Class XII, Electromagnetic Induction and A.C. Fundamentals etc.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Recall** the basic concepts of electrical engineering	**BT 1**
CO 2	**Understand** the concept behind basic electric and magnetic circuits.	**BT 2**
CO 3	**Apply** the working principles of electrical machines in real-life.	**BT 3**
CO 4	**Analyze** DC & AC circuits using different laws and theorems.	**BT 4**

Detailed Syllabus:

Modules	**Topic**	**Course Content**	**Hours**

Modules	Topic	Course Content	Hours
I.	**DC Circuits**	Electrical Circuit Elements – The resistance element, the inductance element, the capacitance element. Voltage & Current source, practical & ideal voltage and current sources, source transformation. Kirchhoff's Laws, Analysis of simple circuits with DC excitation – series circuit, parallel circuit, voltage and current divider rule, star -delta conversion, Maxwells mesh current method, nodal voltage analysis , Network Theorems – Thevenin's Theorem, Nortons Theorem, Superposition theorem	**09**
II.	**AC Circuits**	AC fundamentals – generation of alternating voltage, representation of sinusoidal waveform, concept of frequency, cycle, time period, instantaneous value, average value, peak value, RMS value, phasor representation. Single phase AC Circuits – analysis of single-phase AC circuits consisting of R-L-C parameters, apparent power, real power, reactive power, power factor and its significance. Analysis of R-C series circuit, R-L-C series circuit, analysis of AC parallel circuits.	**09**
III.	**Electrical Machines:**	Principle of operation and construction of single-phase transformers. EMF equation, losses, efficiency and voltage regulation. DC Machines – Constructional details of a DC Machine; EMF Equation of a DC machine, Types of DC Machines, Applications of DC Generators, operation of a DC machine as a motor, Torque equation, importance of back emf, speed equation, speed regulation, starting a DC motor, types of DC Motor, applications of DC motors	**09**
IV.	**Electrical Installations:**	Electrical Power Supply System. Three phase four wire distribution system. Protection of electrical installations against overload, short circuit and earth fault. Protective devices for overload, short circuit, earth fault and electric shock – SFU, MCB, ELCB. Earthing – difference between neutral wire & earth wire, methods of earthing of domestic fittings and appliances. Types of wires, cables and wiring used in electrical installations.	**09**
		TOTAL	**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A Text Book of Electrical Technology*, Thereja, B.L., 1st Edition revised, 2008, S Chand & Company Ltd. Ram Nagar; New Delhi.
2. *Basic Electrical Engineering*, D. P. Kothari, I. J.Nagrath, 3rd Edition, 2009, Tata McGraw-Hill

Reference Books:

1. D. C. Kulshreshtha, *Basic Electrical Engineering*, 1st Edition, 2009, McGraw-Hill
2. E. Hughes, *Electrical and Electronics Technology,* 10th Edition, 2011, Pearson Publication

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Basic%20Electrical%20Engineering%20R-20.pdf
2. https://www.cet.edu.in/noticefiles/231_BASIC_ELECTRICAL_ENGG-min.pdf
3. NPTEL Course on Basic Electrical Circuits by Prof. Nagendra Krishnapura, IITM
4. NPTEL Course on Fundamentals of Electrical Engineering by Prof. Debapriya Das, IIT, Kharagpur

Paper V/Subject Basic Electrical Engineering Lab		**Subject Code: CSE022C113**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make students understand fundamental electrical circuit components, instruments, and their practical use, develop the ability to experimentally verify basic laws and theorems of electrical engineering, enable measurement of electrical parameters (voltage, current, power, energy) using standard equipment, etc.

Prerequisites: Basic concepts of D.C. networks of Class XII, Electromagnetic Induction and A.C. Fundamentals etc.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** and **explain** the function of electrical circuit elements (resistors, capacitors, inductors, meters).	**BT 1 & 2**
CO 2	**Experimentally** verify network theorems (Ohm's Law, KVL, KCL, Thevenin's, Superposition, etc.).	**BT 3**
CO 3	**Analyze** and **interpret** experimental data; compute percentage errors and discuss sources of error.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Lab	**Experiments**	**Hours**
I	To verify Thevenin's Theorem for DC network	**2**
II	To verify Maximum Power Transfer Theorem for DC network	**2**
III	Study of R-L-C Series circuit and determine R,L,C, cos Φ,P and Q and draw the phasor diagram	**2**
IV	Study of R-L-C Parallel circuit and determine R,L,C,cos Φ,P and Q and draw the phasor diagram	**2**
V	Calibration of a milli-ammeter as a voltmeter.	**2**
VI	To determine the ohmic and effective resistance (armature winding)	**2**

VII	To study the characteristics of a filament lamp	**2**
VIII	To measure the power in a single-phase load using one wattmeter	**2**
IX	To measure the insulation resistance using Megger	**2**
X	Demonstration of house wiring	**2**
	TOTAL	**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A Text Book of Electrical Technology*, Thereja, B.L., 1st Edition revised, 2008, S Chand & Company Ltd. Ram Nagar; New Delhi.
2. *Basic Electrical Engineering*, D. P. Kothari, I. J.Nagrath, 3rd Edition, 2009, Tata McGraw-Hill

Reference Books:

1. D. C. Kulshreshtha, *Basic Electrical Engineering*, 1st Edition, 2009, McGraw-Hill
2. E. Hughes, *Electrical and Electronics Technology,* 10th Edition, 2011, Pearson Publication

Paper VI/Subject Name: Engineering Graphics & Design Lab		**Subject Code: CEE022C114**
L-T-P-C – 0-0-4-2	**Credit Units: 02**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make students understand the process of drawing projections and sections and basic engineering drawing formats and to convert sketches to engineered drawings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** with the concepts of drawings and projections	**BT 1**
CO 2	**Understand** the dimension and figures using the drawing instruments and acquire visualization skills, projection of points, etc.	**BT 2**
CO 3	**Utilize** engineering curves in tracing the paths of simple machine components.	**BT 3**

CO 4	**Analyze** and assess sketches to convert them to engineered drawings.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Introduction and Projections**	Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute; Scales – Plain, Diagonal and Vernier Scales; Principles of Orthographic Projections-Conventions - Projections of Points and lines inclined to both planes; Projections of planes inclined Planes - Auxiliary Planes; Covering those inclined to both the Planes- Auxiliary Views; Draw simple annotation, dimensioning and scale. Floor plans that include: windows, doors, and fixtures such as WC, bath, sink, shower, etc	**11**
II.	**Angular Solids and Isometric Projections**	Prism, Cylinder, Pyramid, Cone – Auxiliary Views; Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone; Draw the sectional orthographic views of geometrical solids, objects from industry and dwellings (foundation to slab only). Principles of Isometric projection – Isometric Scale, Isometric Views, Conventions; Isometric Views of lines, Planes, Simple and compound Solids; Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions;	**11**
III.	**Overview of Computer Graphics**	Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects.; Isometric Views of lines, Planes, Simple and compound Solids]. Consisting of set up of the drawing page and the printer, including scale settings, setting up of Modules and drawing limits; ISO and ANSI standards for coordinate dimensioning and tolerancing; Orthographic constraints, Snap to objects manually and automatically; Producing drawings by using various coordinate input entry methods to draw straight lines, Applying various ways of drawing circles;	**11**
IV	**Customisation and CAD drawing**	Covering applying dimensions to objects, applying annotations to drawings; Setting up and use of Layers, layers to create drawings, Create, edit and use customized layers; Changing line lengths through modifying existing lines (extend/lengthen); Printing documents to paper using the print command; orthographic projection techniques; Drawing sectional views of composite right regular geometric solids and project the true shape of the sectioned surface; Drawing annotation, Computeraided design (CAD) software modeling of parts and assemblies. Parametric and non-parametric solid, surface, and wireframe models. Part editing and two-dimensional documentation of models. Planar projection theory, including sketching of perspective, isometric, multiview, auxiliary, and section views. Spatial visualization exercises. Dimensioning guidelines, tolerancing techniques; dimensioning and scale multi views of dwelling;	**11**
		TOTAL	**44**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	2 * 15 NCH = 30 NCH	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Engineering Drawing;* Bhatt, N.D, 53rd Edition, 2016, Charotar Publishing House

Reference Books:

1. Jolhe Dhananjay A; *Engineering drawing*, 5th Edition, 2010, Tata McGraw-Hill Education Pvt. Ltd., New Delhi

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Engineering%20Graphics%20Manual%20final.pdf
2. https://www.pvpsiddhartha.ac.in/autonomus14/1-1/it/IT1L3.pdf
3. NPTEL Course on Engineering Drawing and Computer Graphics by Prof. Rajaram Lakkaraju, IIT, Kharagpur
4. NPTEL Course on Engineering Graphics by Prof. Nihar Ranjan Patra, IIT, Kanpur

Paper VII/Subject Name: Programming for Problem Solving-I		**Subject Code: CSE022C105**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students develop computational thinking and problem-solving skills, learn the fundamentals of C programming, design algorithms and implement solutions using control structures, arrays and strings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the various constructs used in programming	**BT 1**
CO 2	**Explain** the fundamental concepts of programming, algorithms, flowcharts, data types, operators, and control structures in C.	**BT 2**
CO 3	**Develop** C programs using decision-making and iterative constructs to solve computational problems	**BT 3**
CO 4	**Analyze** and debug the errors while writing the programs.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Fundamentals of Programming**	Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.) Idea of Algorithm: steps to solve logical and numerical problems. Representation of Algorithm: Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code.	**15**
II	**C Fundamentals**	Structure of a C program, Character set, tokens, keywords and identifiers, Variables and constants, Data types, Operators and expressions, Input and output functions, Type conversion, Compilation and execution of C programs	**15**
III	**Decision Making and Iterative Programming**	Conditional statements: if, if-else, nested if, else-if ladder, switch. Looping constructs: while, do-while, for, Nested loops, break, continue and goto. Programming using loops, Pattern generation, Number-based applications, Programs based on condition and looping statements	**15**
IV	**Arrays, Strings and Basics of Functions**	One-dimensional arrays: declaration, initialisation, Two-dimensional arrays, Matrix operations, Character arrays, Strings, Standard string library functions, Applications of arrays and strings, Searching in arrays, Sorting techniques, Programming exercises Introduction to User defined functions: function prototype, function call, function declaration, call by value, call by reference	**15**
TOTAL			**60**

Programming for Problem Solving-I Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Character set, Tokens, Keywords and Identifiers, Constants, variables, data types, statements, comments, declaration of storage class, assigning values to variables.
2. Managing I/O, reading and writing characters, formatted Input/output.
3. Arithmetic operators, relational operators, logical operators, assignment operators, increment & decrement operators, conditional operators, bitwise operators, special operators.
4. Importance of decision making, decision making with if statement, if-else statement, nested if-else statements, switch-case statement.
5. Importance of iterative statements, the while statement, do-while statement, for statement, nested for looping.
6. Significance of Arrays, creation and use of one & two-dimensional arrays
7. Declaration and use of string variables, reading and writing strings.
8. User defined functions

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar,*Let us C,* 15th Edition, 2017, BPB.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Programming%20for%20Problem%20Solving.pdf
2. NPTEL course on Introduction to Programming in C by Prof. Satyadev Nandakumar, IIT, Kanpur

Paper IX/Subject Name: Universal Human Values		**Subject Code: BHS022A101**
L-T-P-C – 2-0-0-2	**Credit Units: 02**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to help the students appreciate the essential complementarily between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic need of human values in real life	**BT 1**
CO 2	**Understand** the importance of following the basic universal human values	**BT 2**
CO 3	**Apply** the holistic understanding in one's day-today life so as to keep oneself happy and to socialize with nature, society, etc	**BT 3**
CO 4	**Analyze** the harmony within human beings by distinguishing the needs of the self and the body.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Value Education**	Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education),	**06**

		Understanding Value Education, Sharing about Oneself, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Exploring Human Consciousness, Happiness and Prosperity – Current Scenario, Lectured, Method to fulfil the Basic Human Aspirations, Exploring Natural Acceptance	
II	**Harmony in Human Being**	Understanding Human being as the Co-existence of the Self and the Body, distinguishing between the Needs of the Self and the Body, Exploring the difference of Needs of Self and Body, The Body as an Instrument of the Self Understanding Harmony in the Self, Exploring Sources of Imagination in the Self, Harmony of the Self with the Body, Programme to ensure self-regulation and Health, Exploring Harmony of Self with the Body	**06**
III	**Harmony in the Family & Society**	Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship", Exploring the Feeling of Trust, 'Respect' – as the Right Evaluation, Exploring the Feeling of Respect, Other Feelings, Justice in Human-to-Human Relationship Understanding Harmony in the Society, Vision for the Universal Human Order, Exploring Systems to fulfil Human Goal	**06**
IV	**Harmony in Nature & Implications of Holistic Understanding**	Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature, Exploring the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence, Exploring Co-existence in Existence Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, Exploring Ethical Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics, Exploring Humanistic Models in Education, Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession Exploring Steps of Transition towards Universal Human Order	**06**
TOTAL			**24**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 12 NCH = 24 NCH	-	36 NCH (Seminar, Case Study, Discussion, Internship)

Text Books:

1. *A Foundation Course in Human Values and Professional Ethics,* R R Gaur, R Asthana, G P Bagaria, 2nd Revised Edition, Excel Books, New Delhi

Reference Books:

1. *Human Values*, A.N. Tripathi, 3rd Edition, 2019, New Age Intl. Publishers, New Delhi,

Additional Readings:

1. https://uhv.org.in/uhv2notes
2. https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php

Paper VI/Subject Name: Design Thinking and Ideation Lab		**Subject Code: CSE022S116**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students understand the principles and process of design thinking, develop empathy for users and identify real-world problems, apply creative thinking and ideation techniques to generate innovative solutions, build and test prototypes through iterative design, and work collaboratively to develop user-centric solutions.

Prerequisites: None

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Tell** the utility of design thinking and ideation lab	**BT 1**
CO 2	**Explain** the stages and principles of the design thinking process for solving user-centric problems	**BT 2**
CO 3	**Apply** empathy, observation, and problem-definition techniques to identify real-world challenges.	**BT 3**
CO 4	**Analyze** user requirements and generate innovative solutions using structured ideation techniques.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 22 (2 hours per week)

Minimum 08 Laboratory experiments based on the following-

1. Understanding Design Thinking and Problem Identification: Introduction to Design Thinking and Innovation Process, dentification of real-world problems from society, campus, healthcare, agriculture, education, environment, or industry, Stakeholder mapping and user identification, Conducting interviews, observations, and empathy mapping,
2. Define and Ideate: Problem statement formulation using "How Might We (HMW)" questions, Root cause analysis using Fishbone Diagram and 5-Why Technique, Brainstorming and idea generation techniques, Idea screening using Impact–Feasibility Matrix or Decision Matrix, Storyboarding and user journey mapping.
3. Prototype Development: Low-fidelity prototype development (paper prototype, mock-up, sketch, wireframe, or physical model), Digital prototype creation using appropriate software/tools (e.g., Figma, Canva, Tinkercad, MIT App Inventor, or equivalent), Preparing value proposition and solution canvas, Prototype refinement based on peer feedback.

4. Testing and Project Presentation: User testing and feedback collection, Prototype modification and validation, Preparation of Business Model Canvas (basic introduction), Final team project presentation, demonstration, and documentation.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
-	1 * 22 NCH = 22 NCH	8 * 1 NCH = 8 NCH (Seminar, Case Study, Discussion, Internship)

Text Books:

1. *Developing Thinking Skills (The Way to Success)*, E. Balaguruswamy, 1st Edition, 2022, Khanna Publishing House
2. *Design Thinking for Engineering: A practical guide;* Iñigo Cuiñas, Manuel José Fernández Iglesias, 2023, Institution of Engineering and Technology
3. *Design Thinking For Strategic Innovation: What They Can't Teach You at Business or Design School,* Idris Mootee, 1st Edition, 2014, Adams Media
4. Chris Hackett, *The Big Book of Maker Skills: Tools & Techniques for Building Great Tech Projects,* Reprint Edition, 2018, Weldon Owen
5. Paul Horowitz, Winfield Hill, *The Art of Electronics,* 3rd Edition, 2015, Cambridge University Press

Reference Books:

1. Christian Müller-Roterberg; *Design Thinking For Dummies*, 1st Edition, 2020, For Dummies
2. *A Text Book of DESIGN THINKING For B.TECH. 4th Year, Semester-VII, Suitable For All The 4th Year B-Tech Students*
3. Simon Monk, *Programming Arduino: Getting Started with Sketches*, 2nd Edition, 2016, McGraw Hill TABH
4. Simon Monk, *Make Your Own PCBs with EAGLE: From Schematic Designs to Finished Boards,* 2014, McGraw Hill Education
5. Scott Chacon, Ben Straub, *Pro Git*, 2nd Edition, 2014, A Press
6. Chapman W.A.J, *Workshop Technology*, 5th Edition, 2002, CBS Publishers and distributors

Additional Reading:

1. https://www.aicte-india.org/sites/default/files/IDC/idealab/AICTE%20-%20IDEA%20LAB%20User%20Manual.pdf
2. https://vignaniit.edu.in/ideaLab.php

6.3 Detailed Syllabus of 2nd Semester

Paper I/Subject Name: Chemistry		**Subject Code: CHY022C201**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to s to acquaint the students with the basic phenomenon/concepts of chemistry, the student faces during course of their study in the industry and Engineering field and to understand the new developments and breakthroughs efficiently in engineering and technology

Prerequisites: Concepts of +2 level Chemistry

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic properties of chemical reactions	**BT 1**
CO 2	**Interpret** periodic properties such as ionization potential, electronegativity, oxidation states, electronegativity and bulk	**BT 2**
CO 3	**Experiment with** major chemical reactions that are used in the synthesis of molecules.	**BT 3**
CO 4	**Analyze** microscopic chemistry in terms of atomic and molecular orbitals and intermolecular	**BT 4**

Detailed Syllabus:

Module	**Topics**	**Course Content**	**Periods**
I.	**Atomic and Molecular Structure**	Schrodinger equation. Particle in a box solution and their applications for conjugated molecules and nanoparticles. Forms of the hydrogen atom wave functions and the plots of these functions to explore their spatial variations. Molecular orbitals of diatomic molecules and plots of the multicenter orbitals. Equations for atomic and molecular orbitals. Energy level diagrams of diatomic. Pi-molecular orbitals of butadiene and benzene and aromaticity. Crystal field theory and the energy level diagrams for transition metal ions and their magnetic properties. Band structure of solids and the role of doping on band structures.	**09**
II.	**Spectroscopic Techniques and Applications, Intermolecular Forces and Potential Energy Surfaces**	Principles of spectroscopy and selection rules. Electronic spectroscopy. Fluorescence and its applications in medicine. Vibrational and rotational spectroscopy of diatomic molecules. Applications. Nuclear magnetic resonance and magnetic resonance imaging, surface characterization techniques. Diffraction and scattering. Ionic, dipolar and van Der Waals interactions. Equations of state of real gases and critical phenomena. Potential energy surfaces of H3, H2F and HCN and trajectories on these surfaces.	**09**
III.	**Use of free Energy in Chemical Equilibria and Periodic Properties**	Thermodynamic functions: energy, entropy and free energy. Estimations of entropy and free energies. Free energy and EMF. Cell potentials, the Nernst equation and applications. Acid base, oxidation reduction and solubility equilibria. Water chemistry. Corrosion. Use of free energy considerations in metallurgy through Ellingham diagrams. Effective nuclear charge, penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electronegativity, polarizability, oxidation states, coordination numbers and geometries, hard soft acids and bases, molecular geometries	**09**
IV.	**Stereochemistry, Organic**	Representations of 3 dimensional structures, structural isomers and stereoisomers, configurations and symmetry and chirality,	**09**

	Reactions and Synthesis of a Drug Molecule	enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds Introduction to reactions involving substitution, addition, elimination, oxidation, reduction, cyclization and ring openings. Synthesis of a commonly used drug molecule	
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. *A Textbook of Physical Chemistry*, Negi A.S. and Anand S.C., 2nd Edition, 2007, New Age International
2. *Concise Inorganic Chemistry,* Lee J.D., 5th Edition, 2008, John Wiley and Sons Ltd.

Reference Books:

1. Atkins, P.W. and Paula, J. De,*Physical Chemistry,* 10th Edition, 2014, Oxford University Press
2. Huheey, J.E. Keiter, E.A. Keiter, R.L Medhi, O.K.,*Inorganic Chemistry Principles of Structure and Reactivity*, 4th Edition, 2006, Pearson Education

Additional Readings:

1. Organic Chemistry: Structure and Function by K. P. C. Volhardt and N. E. Schore, 5th Edition, http://bcs.whfreeman.com/vollhardtschore5e/default.asp
2. NPTEL Course on Chemistry - I by Prof. Mangala Sunder Krishnan, IITM

Paper II/Subject Name: Chemistry Lab		**Subject Code: CHY022C211**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to acquaint the students to fundamental experimental techniques and good laboratory practices in chemistry, enable accurate preparation and standardization of chemical solutions, develop skills to perform basic quantitative and qualitative chemical analysis, etc.

Prerequisites: Concepts of +2 level Chemistry

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Identify** and **explain** the purpose of common laboratory equipment and safety protocols.	**BT 1 & 2**

CO 2	**Prepare** standard solutions and perform experiments involving volumetric and instrumental analysis	**BT 3**
CO 3	**Analyze** experimental data to calculate concentration, purity, and properties of given samples.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Determination of surface tension and viscosity.
2. Thin layer chromatography.
3. Ion exchange column for removal of hardness of water.
4. Determination of chloride content of water.
5. Colligative properties using freezing point depression.
6. Determination of the rate constant of a reaction.
7. Determination of cell constant and conductance of solutions.
8. Potentiometry - determination of redox potentials and EMFs.
9. Synthesis of a polymer/drug.
10. Saponification/acid value of an oil.
11. Chemical analysis of a salt.
12. Lattice structures and packing of spheres.
13. Models of potential energy surfaces.
14. Chemical oscillations- Iodine clock reaction.
15. Determination of the partition coefficient of a substance between two immiscible liquids.
16. Adsorption of acetic acid by charcoal.
17. Use of the capillary viscosimeters to the demonstrate of the isoelectric point as the pH of minimum viscosity for gelatin sols and/or coagulation of the white part of egg

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. *A Textbook of Physical Chemistry*, Negi A.S. and Anand S.C., 2nd Edition, 2007, New Age International
2. *Concise Inorganic Chemistry,* Lee J.D., 5th Edition, 2008, John Wiley and Sons Ltd.

Reference Books:

1. Atkins, P.W. and Paula, J. De,*Physical Chemistry,* 10th Edition, 2014, Oxford University Press
2. Huheey, J.E. Keiter, E.A. Keiter, R.L Medhi, O.K.,*Inorganic Chemistry Principles of Structure and Reactivity*, 4th Edition, 2006, Pearson Education

Paper III/Subject Name: Mathematics-II		**Subject Code: MAT022C202**
L-T-P-C – 3-1-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the students Mathematics fundamentals necessary to formulate, solve and analyze engineering problems

Prerequisites: Concepts of Mathematics I

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the methodologies used for solving various equations	**BT 1**
CO 2	**Understand** essential tool of matrices and linear algebra in a comprehensive manner	**BT 2**
CO 3	**Utilize** the essential tools in the field of applied sciences and related fields.	**BT 3**
CO 4	**Analyze** and **evaluate** the qualitative behavior of solutions of systems of differential equations and interpret in the context of an underlying model.	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Matrices** .	Linear Systems of Equations; Linear Independence; Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem; System of linear equations; Symmetric, skew-symmetric and orthogonal matrices; Determinants; Eigenvalues and eigenvectors; Orthogonal transformation; Diagonalization of matrices; Cayley-Hamilton Theorem.	**12**
II	**Differential Equations**	Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type. Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.	**12**
III	**Complex Variable Differentiation**	Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their properties.	**12**
IV	**Complex Variable Integration**	Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour	**12**

TOTAL	48

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A text book of Engineering Mathematics*, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
2. *Mathematical Methods for Physics and Engineering: A Comprehensive Guide,* K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press

Reference Books:

1. Grewal B. S., *Higher Engineering Mathematics,* 43rd Edition, 2014, Khanna Publishers.
2. Raisinghania M.D., *Ordinary and Partial Differential Equations*, 17th Edition, 2014, S. Chand and Co., New Delhi.
3. Narayna S., *A Text Book of Vector Calculus,* Revised Edition, 2009, S. Chand & Co., New Delhi.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/R-18%20Mathematics-II.pdf
2. http://www.bosecuttack.in/studentcorner/LECTURE_NOTE.MATH2.2ND_SEM__1_.pdf
3. https://www.srividyaengg.ac.in/coursematerial/Iyear/111223.pdf

Paper III/Subject Name: Basic Environmental Engineering		**Subject Code: CSE022C203**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

This course focuses on understanding various bio-potentials and their propagation, exploring different types of electrodes and their optimal placement for diverse recordings. It includes the design of bio-amplifiers for physiological recordings and examines measurement techniques for non-physiological parameters. Additionally, it aims to provide familiarity with biochemical measurement methods.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the structure and functions of ecosystems, the importance of biodiversity, and the principles of sustainable development including the UN SDGs.	**BT 1 & 2**
CO 2	**Apply** principles of water and wastewater treatment, solid waste management, and sustainable sludge disposal techniques in environmental engineering contexts.	**BT 3**

CO 3	**Apply** knowledge of air pollution, soil/groundwater contamination, and climate change to suggest appropriate mitigation and remediation strategies.	**BT 3**
CO 4	**Make** decisions using environmental modelling techniques, life cycle assessment (LCA), and risk assessment frameworks to evaluate environmental policies and standards such as ISO 14001.14001.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Ecosystems, Sustainability & Biodiversity**	Introduction to ecology and ecosystem structure and function - Sustainable development principles and UN Sustainable Development Goals (SDGs) - Biodiversity: value, threats, conservation strategies	**15**
II	**Water and Waste Water Treatment**	Principles of water, wastewater, Stormwater and industrial water treatment and management practices, Solid waste management: Sludge handling and sustainable disposal techniques	**10**
III	**Air Pollution**	Air pollutants, sources, transport, and fate, Soil and groundwater remediation: bioremediation, phytoremediation, emerging nanotechnologies, Climate change and global warming: causes and mitigation strategies	**15**
IV	**Environmental Modelling, Policy & Regulation**	Mass balance and life cycle assessment (LCA) in environmental systems, Environmental modelling approaches (numerical/software-based) , Environmental laws and policies: Indian and international (Air Act, Water Act, EPA, Solid Waste Rules), Risk assessment frameworks and ISO standards (e.g., ISO 14001)	**20**
TOTAL			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Biomedical Instrumentation and measurement,* Leslie Cromwell, 2nd Edition, 1990, Prentice Hall of India, New Delhi.
2. *Medical Instrumentation Application and Design,* John G. Webster, 4th Edition, 2009, John Wiley and sons, New York, (Units I, II & V)

Reference Books:

1. Myer Kutz, *Standard Handbook of Biomedical Engineering and Design*, 202, McGraw Hill Publisher
2. Khandpur R.S, *Handbook of Biomedical Instrumentation*, 3rd Edition, 2014, Tata McGraw-Hill, New Delhi, (Units II & IV)
3. Joseph J. Carr and John M. Brown, *Introduction to Biomedical Equipment Technology*, 4th Edition, 2004, Pearson Education.
4. R. Anandanatarajan, *Biomedical Instrumentation*, Kindle Edition, 2015, PHI Learning
5. M. Arumugam, *Biomedical Instrumentation*, 2003, Anuradha Agencies Publishers

Additional Readings:

1. https://www.studocu.com/in/document/aryabhatta-knowledge-university/btechit-btechcse/biology-notes-for-engineers/61016774
2. https://www.aminotes.com/2017/02/biology-for-engineers-module-1-cocepts.html
3. https://topperworld.in/b-tech-biology-notes/

Paper IV/Subject Name: Programming for Problem Solving-II		**Subject Code: CSE022C204**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students develop modular programming skills using functions, understand memory organization through pointers, implement user-defined data types, perform file processing in C.

Prerequisites: Concepts of Programming for Problem Solving-I

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** the concepts of functions, pointers, structures and files in C	**BT 2**
CO 2	**Develop** modular programs using functions and recursion.	**BT 3**
CO 3	**Analyze** and implement programs using pointers and dynamic memory allocation for efficient memory management.	**BT 4**
CO 4	**Design** programs using structures, unions, enumerated data types, and user-defined data types to model real-world problems.	**BT 6**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Concepts of Functions**	Function fundamentals, Function declaration, definition and calling, Function prototypes, Types of functions, Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference. Storage classes, Scope and lifetime of variables, Recursion, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc, programming exercises, applications	**15**

Modules	Topics	Course content	Hours
II	**Pointers and Dynamic Memory Allocation**	Introduction to pointers, Pointer declaration and initialization, Pointer arithmetic, Pointers and arrays, Pointers and strings, Pointers as function arguments, Pointer to pointer, Dynamic memory allocation, malloc(), calloc(), realloc(), free(), Dangling and null pointers, programming exercises, applications	**15**
III	**User Defined Datatypes**	Structures, properties, advantages, initialization, working, Arrays of structures, Nested structures, Passing structures to functions, Pointer to structures, Self-referential structures (concept only), Unions, Enumerated data types (enum, typedef Bit fields (overview), programming exercises, applications	**15**
IV	**File Handling and Preprocessor Directives**	Introduction to files, File opening and closing, Text and binary files, File input/output functions, Random access files, Error handling in files, Command-line arguments, C preprocessor, #include, #define, Macros, Conditional compilation, Multi-file programming (overview), programming exercises, applications	**15**
TOTAL			**60**

Programming for Problem Solving-II Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Programs using user-defined functions
2. Recursive function applications
3. Programs demonstrating storage classes
4. Pointer operations
5. Pointer arithmetic
6. Pointers with arrays
7. Pointers with strings
8. Dynamic memory allocation
9. Structure operations
10. Arrays of structures
11. Nested structures
12. Union applications
13. File read/write operations
14. Random file access
15. Mini project involving modular programming and file handling

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar,*Let us C,* 15th Edition, 2017, BPB.

Additional Readings:

3. https://mrcet.com/downloads/digital_notes/HS/Programming%20for%20Problem%20Solving.pdf
4. NPTEL course on Introduction to Programming in C by Prof. Satyadev Nandakumar, IIT, Kanpur

Paper V/Subject Name: Workshop Practices		**Subject Code: MEE022C215**
L-T-P-C – 0-0-4-2	**Credit Units: 02**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide exposure to the students with hands on experience on various basic engineering practices in Civil, Mechanical, Electrical and Electronics Engineering.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Label** the various techniques used under mechanical engineering	**BT 1**
CO 2	**Understand** the different manufacturing processes which are commonly employed in the industry	**BT 2**
CO 3	**Utilize** tools, instruments and techniques learnt to perform basic household chores in terms of house wiring, carpentry etc.	**BT 3**
CO 4	**Experiment** using the tools and techniques learnt for various purposes and decide on the best prospect.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

The lecture sessions will be on the following topics:

- Manufacturing Methods- casting, forming, machining, joining, advanced manufacturing methods.
- CNC machining, Additive manufacturing.
- Fitting operations & power tools.
- Electrical & Electronics.
- Carpentry.
- Plastic moulding, glass cutting.
- Metal casting.
- Welding (arc welding & gas welding), brazing g topics:

And the lab sessions will on the topics:

- Machine shop

- Fitting shop
- Carpentry
- Electrical & Electronics
- Welding shop (Arc welding + Gas welding)
- Casting
- Smithy
- Plastic moulding & Glass Cutting

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	2*12 NCH = 24 NCH	36 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *Elements of Workshop Technology,* Hajra Choudhury, S K, Hajra Choudhury, A K, 14th Edition, 2007, Mumbai Media Promoters
2. *Manufacturing Technology – I,* Gowri P. Hariharan and A. Suresh Babu, 2008, Pearson Education.

Reference Books:

1. Roy A. Lindberg, *Processes and Materials of Manufacture*", 4th Edition, 1998, Prentice Hall India,

Additional Readings:

1. http://mm-coep.vlabs.ac.in/LaserSpotWelding/Theory.html?domain=Mechanical%20Engineering&lab=Welcome%20to%20Micromachining%20laboratory
2. http://fab-coep.vlabs.ac.in/exp7/Theory.html?domain=Mechanical%20Engineering&lab=Welcome%20to%20FAB%20laboratory

Paper VIII/Subject Name: English for Technical Writing		**Subject Code: CEN022A201**
L-T-P-C – 2-0-0-2	**Credit Units: 02**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide learning environment to practice listening, speaking, reading and writing skills, to assist the students to carry on the tasks and activities through guided instructions and materials and to effectively integrate English language learning with employability skills and training.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the various forms of communication	**BT 1**
CO 2	**Understand** basic proficiency in English.	**BT 2**
CO 3	**Develop** reading and listening comprehension, writing and speaking skills.	**BT 3**
CO 4	**Analyze** the type of communication	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Vocabulary Building**	The concept of Word Formation, Root words from foreign languages and their use in English, Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives, Synonyms, antonyms, and standard abbreviations.	**06**
II	**Basic Writing Skills**	Sentence Structures, Use of phrases and clauses in sentences, Importance of proper punctuation, creating coherence, organizing principles of paragraphs in documents, Techniques for writing precisely, Subject-verb agreement, Noun-pronoun agreement, Misplaced modifiers, Articles, Prepositions, Redundancies, Clichés	**06**
III	**Writing Practices**	Nature and Style of sensible Writing, Describing, Defining 1.3. Classifying, providing examples or evidence, Writing introduction and conclusion, Comprehension, Précis Writing, Essay Writing	**06**
IV	**Oral Communication**	Listening Comprehension, Pronunciation, Intonation, Stress and Rhythm, Common Everyday Situations: Conversations and Dialogues, Communication at Workplace, Interviews, Formal Presentations	**06**
		TOTAL	**24**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2*12 NCH = 24 NCH	--	36 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Effective Communication Skills.* Kul Bhushan Kumar, 2022, Khanna Book Publishing
2. *Practical English Usage,* Michael Swan. 1995, OUP

Reference Books:

1. F.T. Wood, *Remedial English Grammar.,* 2007, Macmillan.
2. William Zinsser, *On Writing Well*, 2001, Harper Resource Book.
3. Liz Hamp-Lyons and Ben Heasly, *Study Writing*, 2006,
4. Sanjay Kumar and PushpLata, *Communication Skills*, 2011, Oxford University Press.

Additional Readings:

1. AICTE's Prescribed Textbook: English (with Lab Manual), Khanna Book Publishing Co., https://khannabooks.com/index.php?route=product/product&path=99_105&product_id=480
2. NPTEL Course on English Language for Competitive Exams by Prof. by Aysha Iqbal, IIT, Madras
3. NPTEL Course on Technical English for Engineers by Prof. by Aysha Iqbal, IIT, Madras

Paper VII/Subject Name: Sports and Yoga		**Subject Code: CSE022C217**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students understand the importance of sound health and fitness principles as they relate to better health, to expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness and to develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Choose** the best form of yoga/ exercise for them	**BT 1**
CO 2	**Understand** basic skills associated with yoga and physical activities including strength and flexibility, balance and coordination	**BT 2**
CO 3	**Experiment with** different forms of yoga to keep oneself physically fit and mentally strong	**BT 3**
CO 4	**Assess** current personal fitness levels	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Physical Education, Olympic Movement, Fitness, Wellness & Lifestyle**	Meaning & definition of Physical Education. Aims & Objectives of Physical Education. Changing trends in Physical Education, Ancient & Modern Olympics (Summer & Winter), Olympic Symbols, Ideals, Objectives & Values, Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc., Meaning & Importance of Physical Fitness & Wellness, Components of Physical fitness, o Components of Health-related fitness, Components of wellness, Preventing Health Threats through Lifestyle Change, Concept of Positive Lifestyle	**03**

II	**Anatomy & Physiology in Physical Education, Sports, Yoga & Postures**	Define Anatomy, Physiology & Its Importance, ffect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.), Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports, o Newton's Law of Motion & its application in sports. o Friction and its effects in Sports, Meaning and Concept of Postures, Causes of Bad Posture. Advantages & disadvantages of weight training. Concept & advantages of Correct Posture. Common Postural Deformities – Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, Bow Legs and Scoliosis. Corrective Measures for Postural Deformities	**03**
III	**Yoga & Lifestyle**	Meaning & Importance of Yoga, Elements of Yoga, Asanas, Pranayama, Meditation & Yogic Kriyas, yoga for concentration & related Asanas (Sukhasana; Tadasana; Padmasana & Shashankasana) Relaxation Techniques for improving concentration - Yog-nidra, Asanas as preventive measures. Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana. Obesity: Procedure, Benefits & contraindications for Vajrasana, Hastasana, Trikonasana, Ardh Matsyendrasana. Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana. Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, Pavan Muktasana, Ardh Matsyendrasana. Asthema: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.	**03**
IV	**Training, Planning and Psychology in Sports**	Meaning of Training, Warming up and limbering down, Skill, Technique & Style, Meaning and Objectives of Planning. Tournament – Knock-Out, League/Round Robin & Combination. Definition & Importance of Psychology in Physical Edu. & Sports, Define & Differentiate Between Growth & Development, Adolescent Problems & Their Management, Emotion: Concept, Type & Controlling of emotions, Meaning, Concept & Types of Aggressions in Sports. Psychological benefits of exercise. Anxiety & Fear and its effects on Sports Performance. Motivation, its type & techniques. Understanding Stress & Coping Strategies	**03**
TOTAL			**12**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
	1*12 NCH = 12 NCH	18 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *Modern Trends and Physical Education*, Ajmer Singh, Gill J.S, Bains J, 4th Edition, 2012, Kalyani Publishers

Reference Books:

1. B.K.S. Iyengar, *Light on Yoga*, 2006, Thorsons

6.4 Detailed Syllabus of 3rd Semester

Paper I/Subject Name: Discrete Mathematics		Subject Code: MAT022C301
L-T-P-C – 3-1-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to make the students learn the concept of mathematical logic, sets, relations, and functions, generating functions and recurrence relations, Graph Theory for solving engineering related problems.

Prerequisites: Concepts of Mathematics I, II

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the concept of logic, sets, relations and functions to solve problems.	**BT 2**
CO 2	**Apply** the concepts learnt to solve computer science related problems.	**BT 3**
CO 3	**Analyze** and evaluate the solutions.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Sets, Relations, Functions & Algebraic Structures**	Operations and Laws of Sets, Binary, Relation, Partial Ordering Relation, Equivalence Relation, Functions, Inverse and Composite Function, Finite and infinite Sets, Countable and uncountable Sets, Poset, Lattice. The Well-Ordering Principle, The Division algorithm: Prime numbers, The Greatest Common Divisor, The least common multiple, Euclidean Algorithm, The Fundamental Theorem of Arithmetic, Congruence, Euler's phi function. Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Congruence Relation and Quotient Structures, Free and Cyclic Monoids and Groups, Permutation Groups, Normal Subgroups, Algebraic Structures with two Binary Operation, Rings, Integral Domain and Fields.	**12**
II	**Graph Theory and Combinatorics**	Graphs and their properties, Degree, subgraphs, walks, paths and circuits, connected and disconnected graphs, Isomorphism, Eulerian and Hamiltonian graphs, Complete graphs, Bipartite graph, Trees, Properties of trees, Pendant vertex, Distance and Centers, Binary tree, Spanning trees, Planar graphs, Matrix representation of graphs, Chromatic number, Chromatic polynomial, Five colours theorem. Pigeon-hole principle, permutation and combination, Recurrence relations, Generating functions.	**12**
III	**Propositional Logic**	Proposition, connectives, tautology, contradiction, logical equivalence, normal forms-DNF, CNF, argument, Validity of	**12**

		argument, fallacy, Rules of Inference, Quantifiers. Boolean Algebra and Boolean Ring, Identities of Boolean Algebra, Duality, Representation of Boolean Function.	
IV	**Probability**	Definition of Probability, Classical, Relative Frequency, and Axiomatic Approaches, Basic Terminologies: Sample Space, Events, Outcomes, Types of Events: Mutually Exclusive, Exhaustive, Independent, Dependent, Addition Theorem of Probability, Conditional Probability, Multiplication Theorem of Probability, Bayes' Theorem and Its Applications, Discrete and Continuous Random Variables, Probability Mass Function (PMF) and Probability Density Function (PDF), Expectation, Variance, and Standard Deviation, Common Discrete Distributions, Joint and Marginal Probability, Independent and Dependent Events, Conditional Independence	**12**
		TOTAL	**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4 * 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A text book of Discrete Mathematics*, Sarkar S. K., Revised Edition, 2016, S Chand & Co Ltd.

Reference Books:

1. Deo N; *Graph Theory with applications to engineering and computer science*, New Edition, 2009, PHI Learning Private Limited.
2. Chandrasekaran N. and Umaparvathi, *Discrete Mathematics*, Eastern Economic Edition, 2013, PHI
3. *Discrete Mathematics and its Applications,* Rosen, K.H., 6th Edition, 2006, McGraw Hill.
4. Tremblay, J.P. and Manohar, R., *Discrete Mathematical Structures with Applications to Computer Science*, 35th Reprint, 2007, Tata McGraw Hill

Paper II/Subject Name: Data Structure & Algorithms		**Subject Code: CSE022C302**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand about the data structures, how to implement them in C, their advantages and drawbacks, & how they can be overcome.

Prerequisites: Concepts of Computer Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy**

		Level
CO 1	**Define** various data structures used in programming.	**BT 1**
CO 2	**Understand** the basic constructs of data structure and its implementation.	**BT 2**
CO 3	**Utilise** the appropriate data structures to solve a given problem.	**BT 3**
CO 4	**Analyse** and **evaluate** the data structures used for problem solving	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Linear Data Structure-I**	a. Introduction: Why we need data structure? Concepts of data structures: Data and data structure, Abstract Data Type and Data Type. Algorithms and programs, basic idea of pseudo-code. Algorithm efficiency and analysis, time and space analysis of algorithms – order notations. b. Array : Different representations – row major, column major. Sparse matrix - its implementation and usage. Array representation of polynomials. c. Linked List: Singly linked list, circular linked list, doubly linked list, linked list representation of polynomial and applications.	**09**
II	**Linear Data Structure-I**	**a. Stack and Queue:** Stack and its implementations (using array, using linked list applications. Queue, circular queue, dequeuers. Implementation of queue- both linear and circular (using array, using linked list), applications. **b. Recursion:** Principles of recursion – use of stack, differences between recursion and iteration, tail recursion. Applications - The Tower of Hanoi, Eight Queens Puzzle.	**09**
III	**Non-Linear Data Structures**	**a. Trees:** Basic terminologies, forest, tree representation (using array, using linked list). Binary trees - binary tree traversal (pre-, in-, post- order), threaded binary tree (left, right, full) - non-recursive traversal algorithms using threaded binary tree, expression tree. Binary search tree- operations (creation, insertion, deletion, searching). Height balanced binary tree – AVL tree (insertion, deletion with examples only). **b. Graphs:** Graph definitions and concepts (directed/undirected graph, weighted/un-weighted edges, sub-graph, degree, cut-vertex/articulation point, pendant node, clique, complete graph, connected components – strongly connected component, weakly connected component, path, shortest path, isomorphism). Graph representations/storage implementations – adjacency matrix, adjacency list, adjacency multi-list. Graph traversal and connectivity – Depth-first search (DFS), Breadth-first search (BFS) – concepts of edges used in DFS and BFS (tree-edge, back-edge, cross-edge, and forward-edge), applications. Minimal spanning tree – Prim's algorithm (basic idea of greedy methods). B-Trees operation	**09**

IV	**Searching and Sorting**	**a. Sorting Algorithms:** Bubble sort and its optimizations, insertion sort, shell sort, selection sort, merge sort, quick sort, heap sort (concept of max heap, application – priority queue), radix sort. **b**. **Searching Algorithms:** Sequential search, binary search, interpolation search.	**09**
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data Structures Using C*, Reema Thareja, 2nd Edition, 2014, Oxford University Press.

Reference Books:

1. Seymour Lipschutz, *Data Structures*, 1st Edition (reprint) 2017, McGraw Hill Education.
2. Narasimha Karumanchi, *Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles*, 5th Edition, 2016, Careermonk Publications.
3. Horowitz, Sahni and Anderson-Freed, *Fundamentals of Data Structures in C*, 2nd Edition, 2008, Universities Press.
4. E. Balagurusamy, *Data Structures Using C*, 1st Edition, 2017, McGraw Hill Education.

Paper III/Subject Name: Data Structure & Algorithms Lab		**Subject Code: CSE022C312**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide the students practical exposure to implement fundamental data structures like stacks, queues, linked lists, trees, and graphs, develop skills to design, implement, and analyze basic and advanced algorithms, enable students to choose appropriate data structures for solving specific computational problems, etc.

Prerequisites: Concepts of Computer Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** the use of recursion and iterative methods in problem solving	**BT 1 & 2**
CO 2	**Implement** and **apply** linear data structures (arrays, linked lists, stacks, queues)	**BT 3**
CO 3	**Analyse** and **evaluate** the data structures used for problem solving	**BT 4 & 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Some common programs of C as revision.
2. Programs on Arrays- Traversal, Insertion, Deletion, Polynomial Representation, etc.
3. Programs on Linked List- Creation Insertion, Deletion, Polynomial Representation, etc.
4. Programs on Stacks-Creation, Push Pop, Infix to Postfix Conversion, Evaluation.
5. Programs on Queues-Creation, Insertion, Deletion, etc.
6. Programs on Trees- Binary Tree Creation, Tree Traversal, BST
7. Programs on Searching- Linear Search, Binary Search
8. Programs on Sorting- Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Heap Sort.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data Structures Using C*, Reema Thareja, 2nd Edition, 2014, Oxford University Press.

Reference Books:

1. Seymour Lipschutz, *Data Structures*, 1st Edition (reprint) 2017, McGraw Hill Education.
2. Narasimha Karumanchi, *Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles*, 5th Edition, 2016, Careermonk Publications.
3. Horowitz, Sahni and Anderson-Freed, *Fundamentals of Data Structures in C*, 2nd Edition, 2008, Universities Press.
4. E. Balagurusamy, *Data Structures Using C*, 1st Edition, 2017, McGraw Hill Education.

Paper IV/Subject Name: Computer Organization and Architecture		**Subject Code: CSE022C303**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain about the machine instructions and basic computer organization and I/O subsystems and pipelining processing

Prerequisites: Fundamental concepts of Digital Logic

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**

CO 1	**Define** the different hardware and its working in a Computer Systems in architectural level	**BT 1**
CO 2	**Demonstrate** computer architecture concepts related to design of modern processors, memories, and I/O	**BT 2**
CO 3	**Solve** problems related to computer Organization and Architecture	**BT 3**
CO 4	**Analyse** the performance of commercially available computers in architectural level.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Hours
I	**Basic organization of computers and machine instructions**	Block level description of the functional units as related to the execution of a program; Fetch, decode and execute cycle. Instruction set architectures, Assembly language programming, addressing modes, instruction cycles, registers and storage, addressing modes; discussions about RISC versus CISC architectures; Inside a CPU.	**09**
II	**Information representation**	Floating point representation (IEEE 754), computer arithmetic and their implementation; Fixed-Point Arithmetic: Addition, Subtraction, Multiplication and Division, Arithmetic Logic Units control and data path, data path components, design of ALU and data path, controller design; Hardwired and Microprogrammed Control	**09**
III	**Memory Technology**	Static and dynamic memory, Random Access and Serial Access Memories, Cache memory and Memory Hierarchy, Address Mapping, Cache updation schemes, Virtual memory and memory management unit.	**09**
IV	**I/O subsystems & Pipeline Processing**	Input-Output devices such as Disk, CD-ROM, Printer etc.; Interfacing with IO devices, keyboard and display interfaces; Basic concepts Bus Control, Read Write operations, Programmed IO, Concept of handshaking, Polled and Interrupt-driven I/O, DMA data transfer. Instruction and Arithmetic Pipeline, Pipeline hazards and their resolution, Parallel Processing.	**09**
		TOTAL	**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Computer System and Architecture*, Moris Mano, 3rd Edition, 2007, PHI.
2. *Structured Computer Organization,* A. S. Tanenbaum, 5th Edition, 2009, Prentice Hall of India

Reference Books:

1. V. C. Hamacher, Z. G. Vranesic and S. G. Zaky, *Computer Organization,* 5th Edition, 20011, McGraw Hill.
2. J. L. Hennessy and D. A. Patterson, *Computer Architecture: A Quantitative Approach,* 5th Edition, 2011, Morgan Kaufmann.

Paper V/Subject Name: Digital Logic and System Design		**Subject Code: CSE022C304**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the simplification of Boolean expression and how to implement with various gates.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the different types of circuits in system design	**BT 1**
CO 2	**Understand** the concepts of combinational and sequential circuit design	**BT 2**
CO 3	**Apply** the concepts learnt to design digital circuits.	**BT 3**
CO 4	**Analyze** the outputs produced and behaviour of the different circuits.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Hours**
I	**Fundamental of Digital Electronics &Boolean algebra and its simplification**	Review of number system; Position number system – decimal, binary, octal and hexadecimal, number base conversion. Representation of negative binary numbers. Codes – BCD Gray, Excess -3. Digital signal, logic gates: AND, OR, NOT, NOR, EX-OR, EX-NOR, Axioms and basic theorem of Boolean algebra. Truth table, logic function and their realization, standard representation (canonical forms) of logic gates-SOP and POS forms, MIN terms and MAX terms Simplification of logic function: K-map of 2, 3, 4 and 5 variables. Simplification of algebra and by map method. Don't care condition. Quine Mcluskey methods of simplification. Synthesis using AND, OR and INVERT and then to convert to NAND or NOR implementation	**09**
II	**Combinational logic circuit design**	Combinational logic circuit and buildings blocks. Binary adders and sub tractors. Carry look ahead adder. Encoders, Decoders, Multiplexers, Demultiplexers, Comparators, parity generators, etc. Realization of logic gates functions through decoders and multiplexers.	**09**

III	**Sequential circuits**	Flip flops: truth table and state table SR, JK, TD, race around condition, master slave conversion of flip-flops. Sequential shift register, sequence generator. Counter s: asynchronous and d Synchronous generators. Ring counter s and Johnson counter, up. Down counter modulo – N counter. Design of Synchronous sequential circuit.	**09**
IV	**Digital logic families and programmable logic devices**	Switching mode operation of PN junction, Bipolar and MOD device Bipolar families: RTL, DTL, DCTL, HTL, TTL, ECL, MOS, and CMOS logic families, Tristate logic. Gate properties fan in, fan out, propagation delay and power delay product. RAM and ROM their uses, SSI, MSI LSI and V LSI devices. Introduction to PLA, PAL TO FPGA and CPLDS, some commonly used digital ICs	**09**
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Digital Logic & Computer Design*, M. Morris Mano, 1st Edition, 2016, Prentice Hall of India.
2. *Digital Principles and Applications*, P. Malvino and D. K. Leach, 8th Edition, 2014, Tata McGraw Hill.

Reference Books:

1. S. Salivahanan and S. Pravin Kumar, *Digital Logic Circuits*, 1st Edition, 2010, Vikas Publishing House.
2. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with VHDL Design*, 3rd Edition, 2017, McGraw Hill.
3. Sanjay Sharma, *Digital Electronics: Digital Logic Design*, 1st Edition, 2013, S K Kataria & Sons.
4. Pratima Manhas and Shaveta Thakral, *Digital Logic & Design*, 1st Edition, 2013, S K Kataria & Sons.
5. A Potton, *An Introduction to Digital Logic*, Imort Edition, 2013, Palgrave.

Paper VI/Subject Name: Digital Logic and System Design Lab		**Subject Code: CSE022C314**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide the students with hands-on experience in designing, implementing, and testing combinational and sequential circuits, enable students to use hardware description languages (HDLs) and digital simulation tools, develop the ability to analyze and troubleshoot digital circuits, etc.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the working of basic logic gates and implement them using discrete components and ICs	**BT 1 & 2**
CO 2	**Design** and **implement** combinational circuits like adders, subtractors, multiplexers, demultiplexers, encoders, and decoders	**BT 3**
CO 3	**Analyze** sequential circuits like flip-flops, counters, and shift registers	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 08 Laboratory experiments based on the following-

- To realize a transistorized AND Gate
- To realize a transistorized OR Gate
- To realize a transistorized NOT Gate
- To realize a transistorized NAND Gate
- To realize a transistorized NOR Gate
- To verify the truth tables of logic gates using ICs
- Realization of half and full adder
- Realization of half and full subtractor
- Realization of 2:1 and 1:2 DEMUX
- Realization of Encoder and Decoder

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Digital Logic & Computer Design*, M. Morris Mano, 1st Edition, 2016, Prentice Hall of India.
2. *Digital Principles and Applications*, P. Malvino and D. K. Leach, 8th Edition, 2014, Tata McGraw Hill.

Reference Books:

1. S. Salivahanan and S. Pravin Kumar, *Digital Logic Circuits*, 1st Edition, 2010, Vikas Publishing House.
2. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with VHDL Design*, 3rd Edition, 2017, McGraw Hill.

Paper VII/Subject Name: OOP using C++		**Subject Code: CSE022C305**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students understand how C++ improves C with object-oriented features and to explain problem solving and programming skills in C++ with extensive programming projects.

Prerequisites: Fundamentals of Computer Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **understand** the basic concepts of OOP.	**BT 1 & 2**
CO 2	**Apply** the concepts learnt to write efficient programs in C++.	**BT 3**
CO 3	**Analyze** a problem and construct a C++ program that solves it.	**BT 4**
CO 4	**Assess** a C++ program and describe ways to improve it.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Introduction, Need, Characteristics, Difference between POP and OOP, Basic concepts of OOP, Features, Applications of OOP. Revision of topics like data types, keywords, identifiers, tokens, reference variables, different operators, conditional and loop control structures.	**15**
II	**Classes and Objects**	Definition of class, object, Difference between class and structure, class definitions, member functions, access specifiers. Objects Dynamic Creation and initialization, Passing and Returning objects, Object assignment and array of objects. Constructors Types, Destructors, Nesting member function, Private member function , Inline functions. Static class members, Function prototyping, Call by reference, Return by reference, Default Argument, Friend functions, this pointer.	**18**
III	**Inheritance and Polymorphism**	Types of Inheritance; Base and Derived classes, Syntax of derived classes, access to the base class; Types of Inheritance, Multiple inheritance, Virtual Base classes, Constructors and Destructors in Inheritance, Container classes, Abstract Classes. Polymorphism: Compile time(Early/Static binding), Overloading functions and operators,Overloading new and delete operators, Run time polymorphism(Late/Dynamic Binding), Virtual functions, Pure Virtual functions, Virtual Destructors, Review of Virtual base classes,	**18**
IV	**Templates, Exception and File Handling**	Templates–Uses, Generic classes, Class templates, Function templates, Advance templates. Examples. Exception handling-Advantages, Try catch and throw clauses, Examples,Manipulators, different examples of manipulators. Pointer types-uses; Dynamic memory allocation techniques, garbagecollection, Linked list, generic pointers; FilesOpen, Close, Read and Write; File attributes, File management	**15**

TOTAL	**66**

OOP using C++ Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Write a C++ program to display "HELLO WORLD".
2. Write a C++ program that will ask the temperature in Fahrenheit and display in Celsius
3. Write a C++ program to print the following output using forloop.

   1
   2 2
   3 3 3
   4 4 4 4
4. Write a C++ program to reverse a number using do-whileloop
5. Write a C++ program to find out the factorial of a number using while loop
6. Write a C++ program to read an integer array and display it.
7. Write a C++ program to read a character array and display it.
8. Write a C++ program to find out the maximum of three number using if-elsestatement
9. Write a C++ program to implement the concept of static data member in class.
10. Write a C++ program to implement the concept of static function in class.
11. Write a C++ program using function with default argument.
12. Write a C++ program to illustrate the use of objects as function arguments (which performs the addition of time in the hour and minutes format)
13. Write a C++ program to illustrate the use of friend function.
14. Write a C++ program to illustrate how an object can be created (within a function) and returned to another function
15. Write a C++ program to illustrate the use of constructors and destructors.
16. Write a C++ program to illustrate the use of copy constructor.
17. Write a C++ program to implement single inheritance (private/public)
18. Write a C++ program to implement multilevel inheritance
19. Write a C++ program to implement multiple inheritances.
20. Write a C++ program to illustrate the use of virtual base class.
21. Write a C++ program to overload unary minus operator
22. Write a C++ program to overload binary „+" operator
23. Write a C++ program to illustrate how an operator can be overloaded using friend function.
24. Write a C++ program to illustrate the use of run time polymorphism.
25. Write a C++ program to swap two variable using function template
26. Write a C++ program to implement try(), catch(), throw()function.
27. Write a C++ program to implement this pointer
28. Write a C++ program to illustrate the use of pointers to derived objects
29. Write a C++ program to illustrate the use of virtual function
30. Write a C++ program to open and close a file using open(), close() function
31. Write a C++ program to illustrate the use of read(), write() function

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Object Oriented Programming With C++,* E. Balaguruswamy, 4th Edition, 2011, Tata McGraw Hill.
2. C++, *The Complete Reference,* Herbert Schildt, 4th Edition, 2017, McGraw Hill Education.

Reference Books:

1. Deital And Deital, *C++ How To Program,* 9th Edition, 2016, Pearson Education India.
2. R. Lafore, *Object Oriented Programming In Turbo C++*, 4th Edition, 2013, Galgotia, New Delhi

6.5 Detailed Syllabus of 4th Semester

Paper I/Subject Name: JAVA Programming		**Subject Code: CSE022C401**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the concepts and implementations of object-oriented programming using JAVA language.

Prerequisites: Basics of Procedural or Object-Oriented Programming

Course Outcomes

On successful completion of the course the students will be able to:

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	**Define** basic structures, constructs, control structure and syntax of JAVA	**BT 1**
CO 2	**Understand** basic idea of object-oriented programming using JAVA	**BT 2**
CO 3	**Apply** the concepts of Java, multithreading and Exception handling to develop efficient and error free codes.	**BT 3**
CO 4	**Analyse** and **evaluate** JAVA programs for its efficiency	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction**	A look at procedure-oriented programming, Object-oriented paradigm, Basic concepts of object-oriented programming (OOP) (encapsulation, inheritance, polymorphism), How Java differs from C and C++, Applications of OOP. Overview of JAVA, Use of math functions, comments, Constructing a java program, Introduction of JVM, Command line argument, Data types, Variables: declaration, scope, Type Conversion and Type Casting, Constants, Operators, Evaluation of Expression, Precedence of Operators, Control statements: selection, iteration, and jump, Introduction to Socket Programming.	**15**
II	**Classes and Objects**	Class: definition and example, Declaring objects, Method overloading and overriding, Binding: concept of binding, static vs. dynamic binding, Using this and super keywords, Access Control, Inheritance: Extending a class, Final, Abstract classes, Constructors Arrays: one-dimensional and multi-dimensional, Strings: string processing functions	**15**
III	**Packages, Interfaces, Exception Handling**	Defining a package, accessing a package and using a package, Interfaces: multiple inheritance, Defining interfaces, implementing interfaces and extending interfaces. Exception handling fundamentals, Exception type: using try…catch, Multiple catch clauses, Throw and Throws Creating threads, Extending the thread class, Stopping and blocking a thread, Life cycle of thread, Threads methods, Thread exceptions, , JDBC,	**15**
IV	**Applets and Files**	Introduction: local and remote applets, How to write applets, Building applet code, Applet life cycle, Creating an executable applet I/O basics, concept of streams, Stream classes: byte stream classes, character stream classes, I/O exceptions, Creation of files, Random access files	**15**
Total			**60**

JAVA Programming Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Write a program in java that outputs your name in giant letters.

- Write a program in Java to find the day of the week of a given date.
- Write a program in Java called Grades Statistics, which reads in n grades (of int between 0 and 100, inclusive) and displays the average, minimum, maximum, and standard deviation.
- Write a program in Java to compute execution time by generating random numbers.
- Write a program in Java to implement the following:
    a. Tokenize the paragraph into single word.
    b. Find the number of word in a paragraph?
    c. Find the number of similar words from the input word.
    d. Find the number of occurrence of each word.
- Write a program in Java to implement some inheritance hierarchy.
- Design and implement an address book object that contains a person's name, home address and phone number, business address and phone number, and numbers for their fax machine, cellular phone, and pager. Write a program in Java to this test plan for the object and implement a driver [finally prepare a package].
- Write a program in Java to demonstrate the use of try, catch, finally throw and throws keywords and demonstrate the following points in the program.
    a. Multiple catch blocks.
    b. try-catch-finally combination.
    c. try-finally combination.
    d. Exception propagation among many methods.
    e. Use of getMessage(), printStackTrace() function of Throwable class.
    f. Nested try blocks
- Write a program that does the following.
    a. Prompts the user for an input file name through a dialog box.
    b. Prompts the user for an output file name through a dialog box.
    c. Copies the input file into the output file, subject to the removal of the space characters listed below from each line.
        i. The leading space characters
        ii. The trailing space characters
        iii. The space characters that are preceded by space characters
- Write a program in Java to design forms.
- Write a program in Java to design a student information form to enter data into the database.
- Write a program in Java to connect some form designed with the back-end database and test them by inserting some records.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 20 NCH = 60 NCH	2 * 15 NCH = 30 NCH	8 * 2 NCH = 16 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Programming with Java: A Primer*; Balagurusamy E., 3rd Edition, 2005, Tata McGraw-Hill, New Delhi
2. *Thinking in Java*, Eckel B., 4th Edition, 2006, PHI.

Reference Books:

1. Maurice N. et al, *Java Generics and Collections*, 1st Edition, 2006, O'REILLY Publication.

2. Booch G., Rumbaugh J. Jacobson I., *The Unified Modeling Language User Guide*, 2nd Edition, 2005, Pearson Education.
3. Schildt H., *The Complete Reference Java,* 7th Edition, 2007, Tata McGraw-Hill, New Delhi

Paper III/Subject Name: Database Management Systems		**Subject Code: CSE022C402**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students learn about databases and the process of designing and constructing data models.

Prerequisites: C/C++, Concepts of Data Structures.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **understand** the basic concepts and applications of database systems	**BT 1 & 2**
CO 2	**Apply** the basic concepts of MySql and write queries using it.	**BT 3**
CO3	**Analyze** the designed database for normalization.	**BT 4**
CO 4	**Evaluate** the process of transaction processing and concurrency control	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Basic Concepts**	Purpose of database systems-Components of DBMS –DBMS Architecture-Three Tier Architecture, and Data Independence-Data modelling -Entity Relationship Model and Diagram, Relational –Network-Hierarchical and object oriented models-Data Modelling using the Entity Relationship Model.	**09**
II	**Structure of Relational Databases**	Relational databases –relational algebra-relational calculus, tuple and domain calculus. Data definition with SQL, insert, delete and update statements in SQL –views –data manipulation with SQL. assertions –triggers, Cursors	**09**
III	**Database Design**	Design guidelines–Relational database design –Integrity Constraints –Domain Constraints-Referential integrity –Functional Dependency-Normalization using Functional Dependencies, Normal forms based on primary keys-general definitions of Second and Third Normal Forms. Boyce-Codd Normal Form–Multi-valued Dependencies and Forth Normal Form –Join Dependencies and Fifth Normal Form –Pitfalls in Relational Database Design, Properties of Relational Decomposition, Dependency Preserving Property, Lossless Non-Additive Join Property, Testing Relational Decompositions for non-additive and dependency preserving properties.	**09**

IV	**Introduction to Transaction and Query Processing**	Transaction and System Concepts-Desirable properties of Transactions-Schedules and Recoverability-Serializability of Schedules -Concurrency Control–Data Storage Indexing and Query processing and Optimization MySQL case study: The basic structure of the MySQL system database structure and its manipulation in MySQL -storage organization in MySQL -Programming in PL/SQL-Cursor in PL/SQL	**09**
		TOTAL	**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Fundamentals of Database System,* Elmasri and Navathe, 7thEdition, 2016, Pearson Education Asia
2. *Database System Concepts*, Henry F Korth, Abraham Silberschatz, 6th Edition, 2013, Mc Graw Hill.
3. *DataBase Management System*, Paneerselvam, 2nd Edition, 2011, PHI Learning

Reference Books:

1. C. J. Date*, An Introduction to Database Systems,* 8th Edition, 2003, Pearson Education Asia
2. Bibin C. Desai, *An Introduction to Database Systems*, Revised Edition, 2012, Galgotia Publications

Paper IV/Subject Name: Database Management Systems Lab		**Subject Code: CSE022C412**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide practical exposure to designing and creating relational databases, enable students to apply MySQL for data definition, manipulation, and control, develop skills to write complex queries, sub-queries, joins, and views, etc.

Prerequisites: C/C++, Concepts of Data Structures.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **understand** the basic concepts and applications of database systems	**BT 1 & 2**
CO 2	**Apply** the basic concepts of MySql and write queries using it.	**BT 3**
CO3	**Analyze** the designed database for normalization.	**BT 4**

CO 4	**Evaluate** the process of transaction processing and concurrency control	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Programs to understand the functionality and limitations of file system.
2. Consider the following relational schema
   Employee (Emp_no, Name, Salary, design, dept_id, DOJ)
   Department (Dept_id, DName, loc, DOE)
   a. Display the name of the employees working in marketing dept.
   b. Display the details of the employee joined in the month of July.
   c. Display the details of the employee who gets maximum salary.
   d. Count the no of employees in each department
3. Consider the following relational schema
   Student (Rollno, Name, Address, DOB, C_id)
   Course (C_id, Cname, Dur, Fees)
   a. Display rollno,name,cname,fees of each student
   b. Count the no of students in each course
4. Consider the following relational schema
   Books(book_id,b_name,author,purchase_date,cost)
   Members(member_id,m_name,address,phone,birthdate)
   Issue_return(book_id,member_id,issue_date,return_date)
   a. Find the author of the books that have not been issued.
   b. Display the member_id and no of books issued to that (Assume that if a book in Issue_Return relation does not have a return_date then it is issued)
   c. Find the book that has been issued the minimum no of times.
   d. Display the names and author of the books that have been issued at any time to a member whose name begins with "Ra".
   e. Display the name and Cost of those books that have been issued to any member whose date of birth is less than 01-01-1989 but not been issued to any member having the birth date equal to or greater than 01-01-1989.
5. Consider the following relational schema
   Student(name,phone,dob,s_id)
   Course(c_id,cname,credit,teacher_id)
   Result(s_id,c_id,mark)
   a. Find the name of the students whose results are not declared in any course
   b. Find the teachers who are teaching more than one course
   c. Display the name and marks of those students who were born before 1-1-1989 and score more than 80 marks in any course
   d. Find the details of students securing pass marks in more than 3 course
   e. Find the total no of credits earned by a students whose id is 10.
   f. Find name of the students who got maximum overall marks.
   g. Display the name and marks of those students who scored more than 80 marks in any subject.
   h. Find the details of the students securing less than 30 marks in more than 3 subjects.
6. Consider the following relational schema
   Customer(C_id, Name , Address)
   Item(i_code , Name , Price)
   Purchase (P_id ,C_id , I_code, qty , pdate)
   a. Find the name of the customer who has done maximum purchase.
   b. Display the name of the item that has been purchased maximum no of times in the month of Feb.
   c. Display the name of the customer who didn't purchase any item.
7. Create three triggers (insert, delete and update) on emp table so that:

a. Whenever a new record is inserted then the emp_id and date of insertion is stored in another table called new_rec.
b. Whenever a record is deleted the emp_id and date of deletion is stored in another table called old_rec.
c. Whenever employee's salary is updated the emp_id , old salary and updated salary is stored in another table called update_info.

8. Write a procedure to accept a emp_id and display the employee details.
9. Write a procedure to accept a emp_id and return the employee salary.
10. Given,
    Emp(emp_no,name,salary,supervisor_no,dept_code)
    Dept(dept_code, dept_name)
    a. employees who get more salary than their supervisor
    b. Department name and total number of employees in each Department.
    c. Name and department of employee(s) who earn maximum salary.
11. Programs on Views and Cursors

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Fundamentals of Database System,* Elmasri and Navathe, 7thEdition, 2016, Pearson Education Asia
2. *Database System Concepts*, Henry F Korth, Abraham Silberschatz, 6th Edition, 2013, Mc Graw Hill.
3. *DataBase Management System*, Paneerselvam, 2nd Edition, 2011, PHI Learning

Reference Books:

1. C. J. Date*, An Introduction to Database Systems,* 8th Edition, 2003, Pearson Education Asia
2. Bibin C. Desai, *An Introduction to Database Systems*, Revised Edition, 2012, Galgotia Publications

Paper V/Subject Name: Formal Language and Automata Theory		**Subject Code: CSE022C403**
L-T-P-C – 3-1-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to impart knowledge on regular grammars, regular expressions and to teach about the basics of parsing and ambiguity.

Prerequisites: Fundamentals of Set Theory

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** and **understand** the utility and importance of Automata Theory as the basis of all computer science languages design	**BT 1 & 2**

CO 2	**Construct** minimized sample automata and grammars of context free languages	**BT 3**
CO3	**Analyze** the power and limitation of a computer and solve the problems using formal language	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction**	Basics of Strings and Alphabets, DFA, transition graphs, regular languages, non-deterministic FA, equivalence of DFA and NDFA	**12**
II	**Grammars**	Regular grammars, regular expressions, equivalence between regular languages, properties of regular languages, pumping lemma.	**12**
III	**Deterministic and Non-Deterministic PDA**	Leftmost and rightmost derivation, parsing and ambiguity, ambiguity in grammar and languages, normal forms. NDPDA, DPDA, context free languages and PDA, comparison of deterministic and non-deterministic versions, closure properties, pumping lemma for CFL	**12**
IV	**Turing Machine**	Turing Machines, variations, halting problem, PCP Chomsky Hierarchy Manipulators, different examples of manipulators; Pointer types- uses; Dynamic memory allocation techniques - garbage collection, Linked list, generic pointers; Files- Open, Close, Read and Write; File attributes, File management	**12**
		TOTAL	**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum/ Tutorial**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *An Introduction to Formal Languages and Automata*, Peter Linz, 3rd Edition, 2010, Narosa Publishers

Reference Books:

1. J. E. Hopercroft and J. D. Ullman,*Introduction to Automata Theory, Languages & Computation*, 3rd Edition, 2006, Narosa
2. J. C. Martin, *Introduction to Languages and The Theory of Computation,* 3rd Edition, 2009, McGraw Hill International Edition.

Paper VI/Subject Name: Mathematical Foundations of AI		**Subject Code: ARI022C405**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to develop a strong mathematical foundation for understanding AI and Machine Learning concepts.

Prerequisites: Basic knowledge of Linear Algebra, Probability, and Calculus

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic mathematical foundations needed for dealing with AI systems	**BT 1**
CO 2	**Understand** probability and statistics to model uncertainty in AI systems.	**BT 2**
CO 3	**Apply** linear algebra concepts to AI applications like dimensionality reduction and embeddings.	**BT 3**
CO 4	**Apply** graph theory for solving AI-related problems such as network analysis and search algorithms.	**BT 3**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Linear Algebra for AI**	Vector Spaces and Matrices: Basis, Span, Rank, Null Space, Orthogonality Matrix Operations: Transpose, Inverse, Determinants, Eigenvalues & Eigenvectors, Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) Norms, Projections, and Distance Metrics Applications in AI: Feature transformation, Dimensionality Reduction, Word Embeddings	**09**
II	**Probability and Statistics for AI**	Probability Theory: Random Variables, Probability Distributions (Bernoulli, Binomial, Gaussian), Bayesian Inference: Bayes Theorem, Maximum Likelihood Estimation (MLE), Maximum A Posteriori (MAP) Expectation, Variance, Covariance, and Correlation, Markov Chains and Hidden Markov Models (HMM) Applications in AI: Naïve Bayes Classifier, Probabilistic Graphical Models, Decision Trees	**09**
III	**Optimization Techniques**	Convex Optimization: Gradient Descent, Stochastic Gradient Descent (SGD), Newton's Method, Lagrange Multipliers and Constrained Optimization, Linear and Quadratic Programming Backpropagation and Optimization in Neural Networks Applications in AI: Training Machine Learning Models, Regularization Techniques	**09**
IV	**Graph Theory and Advanced Topics**	Graph Basics: Graph Representation, Adjacency Matrices, Graph Traversal (BFS, DFS) Shortest Path Algorithms: Dijkstra's Algorithm, Bellman-Ford Algorithm Spectral Graph Theory and Laplacian Matrices Game Theory and Decision Making in AI Applications in AI: Social Network Analysis, Knowledge Graphs, Graph Neural Networks	**09**
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Mathematics for Machine Learning,* Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, 2020, Cambridge University Press
2. *Pattern Recognition and Machine Learning,* Christopher M. Bishop, 1st Edition, 2009, Springer
3. *Probability, Statistics and Random Processes,* T. Veerarajan, 3rd Edition, 2017, McGraw Hill

Reference Books:

1. S. Kumaresan, *Linear Algebra and its Applications*, New Title Edition, 2000, Prentice Hall India
2. N. Deo, *Graph Theory and its Applications,* New Edition, 1979, PHI Learning
3. S. S. Rao, *Optimization Theory & Applications*, 1984, New Age International
4. J. N. Kapur, H. C. Saxena, *Mathematical Statistics*, 2010, S. Chand & Co.

Paper V/Subject Name: Programming with Python		**Subject Code: CSE022G405**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the students about Programming with Python and use it to solve real world problems.

Prerequisites: Fundamentals of Computers

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** semantics and syntax of Typescripts.	**BT 1**
CO 2	**Understand** static types and know how to port untyped JavaScript	**BT 2**
CO 3	**Apply** the concepts learnt to create Single Page Web Applications (SPA) using React, Typescript and Tailwind CSS.	**BT 3**
CO 4	**Inspect** different elements of front-end development	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**

I	**Introduction to Python**	Introduction to Python Programming: Python interpreter/shell, indentation; identifiers and keywords; literals, numbers, and strings; operators (arithmetic operator, relational operator, Boolean operator, assignment, operator, ternary operator and bitwise operator) and expressions	**18**
II	**Programming With Python**	Input and output statements, defining functions, control statements (conditional statements, loop control statements, break, continue and pass, exit function.), default arguments,	**18**
III	**Python Functions and Strings**	Python Functions, Python Lambda, Python Arrays, Python Classes/Object, Inheritance, Iterator, Polymorphism, Scope, Modules, Dates, Maths, JSON,RegEx, PIP, User Input , Strings	**15**
IV	**Python Modules**	Introduction to Numpy, Pandas,SciPy, Django	**15**
Total			**66**

Introduction to Python Programming Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. **Hello World Program:**
   - Write a simple Python program to print "Hello, World!" to the console.
2. **Variable Declaration and Printing:**
   - Practice declaring variables of different types (int, float, string) and printing their values.
3. **Basic Arithmetic Operations:**
   - Write Python scripts to perform basic arithmetic operations such as addition, subtraction, multiplication, and division.
4. **Conditional Statements:**
   - Create programs using if-else statements to perform tasks based on certain conditions.
5. **Loops (for and while):**
   - Practice writing for and while loops to iterate over sequences or execute code repeatedly.
6. **Lists and List Operations:**
   - Explore lists in Python and perform operations like appending, removing, and accessing elements.

   Level: Intermediate
7. **Functions:**
   - Define and call functions to encapsulate reusable code blocks. Practice passing arguments and returning values from functions.
8. **String Manipulation:**
   - Work on tasks involving string manipulation, such as concatenation, slicing, and searching.
9. **File Handling:**
   - Write Python scripts to read from and write to files. Practice handling exceptions during file operations.
10. **Dictionaries and Sets:**
    - Experiment with dictionaries and sets in Python. Perform operations like adding, removing, and accessing elements in dictionaries and sets.
11. **Object-Oriented Programming (OOP) Concepts:**

- Introduce students to OOP concepts like classes, objects, inheritance, and polymorphism. Have them implement simple classes and explore inheritance hierarchies.

12. **Exception Handling:**
    - Practice handling exceptions using try-except blocks to gracefully manage errors in Python programs.
13. **Data Structures and Algorithms:**
    - Implement common data structures (e.g., stacks, queues, linked lists) and algorithms (e.g., sorting, searching) using Python.
14. **Regular Expressions:**
    - Introduce regular expressions and their usage in Python for pattern matching and text processing tasks.
15. **Modules and Packages:**
    - Explore the concept of modules and packages in Python. Have students create their own modules and packages and import them into other scripts.
16. **GUI Programming with Tkinter:**
    - Introduce GUI programming using Tkinter. Have students create simple graphical user interfaces (GUIs) for basic applications.

National Credit Hours		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	--	3*8 = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Textbooks:

1. Guttag, J.V. (2016). Introduction to computation and programming using Python. 2nd edition. MIT Press.

Reference Books:

1. Kamthane, A. N., & Kamthane, A.A. (2017) Programming and Problem Solving with Python, McGraw Hill Education.
2. Liang, Y. D. (2013). Introduction to Programming using Python. Pearson Education.

6.6 Detailed Syllabus of 5th Semester

Paper I/Subject Name: Operating Systems		**Subject Code: CSE022C501**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamental concepts and design of operating systems, apply principles of concurrency, synchronization, and deadlock in operating systems.

Prerequisites: Computer Programming, Computer Architecture & Organization

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy**

		Level
CO 1	**Describe** and **illustrate** the role and responsibilities of an operating system, system calls, kernel vs. user mode, etc.	**BT 1 & 2**
CO 2	**Apply** process scheduling algorithms (FCFS, SJF, RR, etc.) to compute performance metrics.	**BT 3**
CO 3	**Analyze** synchronization problems and solutions like semaphores, monitors, and deadlock.	**BT 4**
CO 4	**Evaluate** different memory management strategies and their impact on performance.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Operating Systems Overview**	Introduction and history of Operating systems, structure and operations; processes and files. **Computer System Overvie**w- Basic Elements, Instruction Execution, Interrupts Memory Hierarchy, Cache Memory, Direct Memory Access, Multiprocessor and Multicore Organization. **Operating system overview**-objectives and functions, Evolution of Operating System.- Computer System Organization- Operating System Structure and Operations- System Calls, System Programs, OS Generation and System Boot	**09**
II	**Process Management And Concurrency Control**	**Processes**-Process Concept, Process Scheduling, Operations on Processes, Interprocess Communication; Threads- Overview, Multicore Programming, Multithreading Models; Thread and SMP Management. **Process Synchronization** – Critical Section Problem, Mutex Locks, Semaphores, Monitors; CPU Scheduling and scheduling algorithms. **Deadlocks**- Shared resources, resource allocation and scheduling, resource graph models, deadlock detection, deadlock avoidance, deadlock prevention algorithms	**09**
III	**Storage Management**	Memory Management requirements, Memory partitioning: Fixed and Variable Partitioning, Memory Allocation: Allocation Strategies (First Fit, Best Fit, and Worst Fit), Fragmentation, Swapping, and Paging. Segmentation, Demand paging, Virtual Memory: Concepts, management of VM, Page Replacement Policies (FIFO, LRU, Optimal, Other Strategies), Thrashing. 32 and 64 bit architecture Examples; Allocating Kernel Memory, OS Examples	**09**
IV	**I/O and File Systems**	I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering, Overview of mass storage structure- disks and tapes. Disk structure – accessing disks, Swap Space. Disk Scheduling (FCFS, SCAN, C-SCAN, SSTF), RAID, Disk Cache. Disk Protection– Goals, Principles, Domain. **File System Interface**: File Concepts – Attributes – operations – types – structure – access methods. File system mounting. Protection. File system implementation. Directory implementation – allocation methods. Free space Management.	**09**
		TOTAL	**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Operating System Concepts,* Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, 9th Edition, 2012, John Wiley and Sons Inc.

Reference Books:

1. William Stallings, *Operating Systems – Internals and Design Principles*, 7th Edition, 2011, Prentice Hall.
2. Andrew S. Tanenbaum, *Modern Operating Systems*, 2nd Edition, 2001, Addison Wesley.
3. D M Dhamdhere, *Operating Systems: A Concept-Based Approach*, 2nd Edition, 2007, Tata McGraw-Hill Education.

Paper II/Subject Name: Operating Systems Lab		**Subject Code: CSE022C511**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide hands-on experience in implementing core operating system concepts like process scheduling, memory management, and synchronization, enable students to develop programs simulating OS functionalities such as CPU scheduling algorithms and deadlock handling, etc.

Prerequisites: Computer Programming, Computer Architecture & Organization

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **illustrate** the role and responsibilities of an operating system, system calls, kernel vs. user mode, etc.	**BT 1 & 2**
CO 2	**Apply** process scheduling algorithms (FCFS, SJF, RR, etc.) to compute performance metrics.	**BT 3**
CO 3	**Analyze** synchronization problems and solutions like semaphores, monitors, and deadlock.	**BT 4**
CO 4	**Evaluate** different memory management strategies and their impact on performance.	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Basic Linux Commands and Overview.
2. Write Shell Script for followings
    - To find the global complete path for any file.
    - To broadcast a message to a specified user or a group of users logged on any terminal.
    - To copy the file system from two directories to a new directory in such a way that only the latest file is copied in case there are common files in both the directories.
    - To compare identically named files in two different directories and if they are same, copy one of them in a third directory
    - To delete zero sized files from a given directory (and all its sub- directories).
    - To display the name of those files (in the given directory) which are having multiple links.
    - To display the name of all executable files in the given directory.
    - Write a script to display the date, time and a welcome message (like Good Morning etc.). The time should be displayed with "a.m." or "p.m." and not in 24 hours notation.
    - Write a script to display the directory in the descending order of the size of each file.
3. Implementation of System Calls.
4. Implementation of FCFS (First Come First Serve) CPU Scheduling.
5. Implementation of SJF (Shortest Job First) CPU Scheduling.
6. Implementation of Round Robin (RR) CPU Scheduling.
7. Implementation of Priority CPU Scheduling Algorithm.
8. Implementation of FIFO Replacement Algorithm.
9. Implementation of Optimal Page Replacement Algorithm.
10. Implementation of LRU Page Replacement Algorithm by Stack method
11. Implement the producer-consumer problem using threads.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Operating System Concepts,* Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, 9th Edition, 2012, John Wiley and Sons Inc.

Reference Books:

1. William Stallings, *Operating Systems – Internals and Design Principles*, 7th Edition, 2011, Prentice Hall.
2. Andrew S. Tanenbaum, *Modern Operating Systems*, 2nd Edition, 2001, Addison Wesley.

Paper III/Subject Name: Data Communication		**Subject Code: CSE022C504**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach about the fundamental principles and models of data communication, apply knowledge of modulation, multiplexing, and switching techniques, etc.

Prerequisites: Concepts of Computer Fundamentals, Digital Logic and Design.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the basic components and types of data communication systems and data transmission concepts.	**BT 1 & 2**
CO 2	**Apply** error detection and correction techniques such as CRC and Hamming code.	**BT 3**
CO 3	**Analyze** data link layer protocols such as HDLC, Stop-and-Wait, and Sliding Window	**BT 4**
CO 4	**Evaluate** performance issues in flow control, multiplexing, and switching techniques.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Introduction**	Introduction to Computer network, Networks: classification and components, Layered architecture of a network software, OSI and TCP/IP model. Data Transmission: Communication model- Simplex, half duplex and full duplex transmission - Periodic Analog signals: Sine wave, phase, wavelength, time and frequency domain, bandwidth – Digital Signals; Digital data Transmission:- Analog & Digital data, Analog & Digital signals, Analog &Digital transmission – Transmission Impairments: Attenuation, Delay distortion, Noise – Channel capacity:	**09**
II	**Signal Analysis**	Introduction to Signal and its Classification. System and its Basic Properties, Spectral Analysis of a Signal; Signal Bandwidth. Transmission media - Guided Transmission Media: Twisted pair, Coaxial cable, optical fiber, Wireless Transmission, Terrestrial microwave, Satellite microwave. Wireless Propagation: Ground wave propagation, Sky Wave propagation, LoS Propagation.	**09**
III	**Data Encoding and Multiplexing**	Baseband Communication: Data Encoding and Modulation, Analog Modulation: AM, FM and PM, Pulse Modulation System: PAM and PWM. Digital Modulation: ASK, FSK, PSK and QAM Multiplexing and its Application: Frequency Division Multiplexing, Wavelength Division Multiplexing, Time Division Multiplexing, Spread Spectrum.	**09**
IV	**Switching and Information Theory and Coding**	Switching: Switching and its Application, Circuit Switching and Packet Switching, Datagram Switching and Virtual Circuit SwitchingX.25, Frame Relay and ATM. Introduction to Information Theory and Average Information Source Coding: Huffman Coding, Error Detection and Correction Codes, Hamming Distance, Linear Block Coding, Cyclic Codes, CRC, Convolution Codes	**09**

TOTAL	36

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data Communications and Networking*, Forouzan B. A., 4th Edition, 2007, Tata McGraw Hill.

Reference Books:

1. Tanenbaum A. S. and D. Wetherall, *Computer Networks*, 5th Edition, 2013, Pearson Education.
2. William Stallings, *Data and Computer Communication,* 9th Edition, 2011, Pearson Education, Inc.

Paper V/Subject Name: Design and Analysis of Algorithms		**Subject Code: CSE022C503**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to enable the students analyze performance of algorithms and solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking and branch and bound.

Prerequisites: Concepts of Data Structures and Basic Mathematics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Examine** and **understand** the performance of algorithm.	**BT 1 & 2**
CO 2	**Apply** different designing methods for development of algorithms to realistic problems, such as divide and conquer, greedy and etc.	**BT 3**
CO 3	**Analyze** and **evaluate** algorithms to improve their efficiency.	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours

I	**Introduction and Divide and Conquer**	Algorithm, Psuedo code for expressing algorithms, Performance Analysis-Space complexity, Time complexity, Asymptotic Notation- Big oh notation, Omega notation, Theta notation and Little oh notation, Probabilistic analysis, Amortized analysis. Master's Theorem	**09**
II	**Searching and Traversal Techniques**	Efficient non - recursive binary tree traversal algorithm, Disjoint set operations, union and find algorithms, Spanning trees, Graph traversals - Breadth first search and Depth first search, AND / OR graphs, game trees, Connected Components, Bi - connected components. Disjoint Sets- disjoint set operations, union and find algorithms, spanning trees, connected components and bi-connected components.	**09**
III	**Types of Problem Solving Techniques**	Greedy Method: General method, applications - Job sequencing with dead lines, 0/1 knapsack problem, Minimum cost spanning trees, Single source shortest path problem. Dynamic Programming: General method, applications-Matrix chain multiplication, Optimal binary search trees, 0/1 Knapsack problem, All pairs shortest path problem,Travelling sales person problem, Reliability design. Backtracking: General method, applications-n-queen problem, sum of subsets problem, graph coloring, Hamiltonian cycles. Branch and Bound: General method, applications - Travelling sales person problem,0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution	**09**
IV	**NP-Hard and NP-Complete Problems**	NP-completeness – Polynomial time verification – Theory of reducibility – Circuit satisfiability - NP-completeness proofs – NP-complete problems: Vertex cover, Hamiltonian cycle and Traveling Salesman problems – Approximation Algorithms – Approximation algorithms to vertex-cover and traveling salesman problems.	**09**
TOTAL			**36**

Design and Analysis of Algorithms Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Implementation and analysis of iterative and recursive algorithms.
- Implementation of searching algorithms: Linear Search, Binary Search
- Implementation of sorting algorithms: Bubble Sort, Selection Sort, Insertion Sort
- Implementation of efficient sorting algorithms: Merge Sort, Quick Sort, Heap Sort
- Implementation of Strassen's method
- Implementation of programs using dynamic programming.
- Implementation of programs using greedy technique.
- Implementation of programs using backtracking technique.
- Implementation of programs using Branch and Bound technique.
- Implementation of graph-related algorithms.
- Implementation of programs using randomized technique.

Credit Distribution		
Lecture/ Tutorial	**Practicum/ Tutorial**	**Experiential Learning**
3* 15 NCH = 45 NCH	1*30 NCH = 30 NCH	45 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Introduction to Algorithms,* T. H. Cormen, C. E. Leiserson, R. L. Rivest, 3rd Edition, 2009, The MIT Press, Cambridge, Massachusetts.

Reference Books:

1. Aho, Hopcroft & Ullman, *The Design and Analysis of Algorithms,* 1974,Addison- Wesley
2. Horowitz & Sahani, *Fundamentals of Algorithms,* 2nd Edition, 2009, Galgotia Publications

Paper VI/Subject Name: Introduction to Artificial Intelligence		**Subject Code: ARI022C505**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to provide undergraduate students with an in-depth understanding of Artificial Intelligence (AI), focusing on core concepts, mathematical foundations, algorithms, and real-world applications.

Prerequisites: Strong foundation in programming (Python recommended), Linear Algebra, Probability & Statistics, and Data Structures & Algorithms

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** understanding of fundamental AI principles, including search algorithms and knowledge representation	**BT 1 & 2**
CO 2	**Implement** AI search techniques, constraint satisfaction problems, and optimization techniques	**BT 3**
CO 3	Design and **analyze** AI models for reasoning, learning, and decision-making	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Introduction to AI**	Definition and history of AI, AI applications, Turing Test, Rational Agents, Search strategies (uninformed: BFS, DFS; informed: A*, Iterative Deepening, Hill Climbing),	**8**

		Constraint Satisfaction Problems (CSP)	
II	**Knowledge Representation and Reasoning**	Logic-based AI (Propositional & First-Order Logic), Rule-based systems, Bayesian Networks, Markov Decision Processes (MDP), Game Theory (Minimax, Alpha-Beta Pruning)	**12**
III	**Machine Learning and Neural Networks**	Supervised vs. Unsupervised Learning, Classification and Regression, Decision Trees, Naïve Bayes, SVMs, Neural Networks (Backpropagation, CNNs, RNNs), Introduction to Reinforcement Learning	**14**
IV	**AI Applications & Advanced Topics**	Natural Language Processing (NLP), Computer Vision, AI in robotics and autonomous systems, Deep Reinforcement Learning, Ethical considerations and bias in AI, Security in AI (adversarial attacks, fairness, explainability)	**14**
Total			**48**

Introduction to Artificial Intelligence Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Introduction to Python libraries for Artificial Intelligence.
- Design and implementation of a simple intelligent agent.
- Implementation of Breadth First Search (BFS).
- Implementation of Depth First Search (DFS).
- Implementation of Best First Search.
- Implementation of A* Search Algorithm.
- Solving Constraint Satisfaction Problems (CSPs) such as Map Coloring or N-Queens.
- Representation of knowledge using propositional logic.
- First Order Logic (FOL) implementation using rule-based systems.
- Development of an expert system using IF-THEN rules.
- Bayesian Network construction and probabilistic reasoning.
- Implementation of Minimax algorithm for game playing.
- Alpha-Beta pruning for optimization of game trees.
- Data preprocessing and feature engineering for AI applications.
- Implementation of Decision Tree classifier.
- Implementation of Naïve Bayes classifier.
- Support Vector Machine (SVM) classification.
- Development of a Single Layer Perceptron.
- Implementation of Multilayer Perceptron using TensorFlow/Keras.
- Comparative analysis of AI and Machine Learning approaches.
- Sentiment Analysis using Natural Language Processing.
- Text Classification using NLP techniques.
- Image Classification using Computer Vision libraries.
- Face Detection/Object Detection using OpenCV.
- Development of a simple Reinforcement Learning environment.
- Mini Project: AI-based application using search, reasoning, learning, or perception techniques.

National Credit Hours		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**

3 * 15 NCH = 45 NCH	1*30= 30 NCH	45 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbook:

1. *Artificial Intelligence: A Modern Approach*, Stuart Russell, Peter Norvig 4th Edition, 2020, Pearson

Reference Books:

1. Elaine Rich, Kevin Knight, Shivashankar B. Nair, *Artificial Intelligence*, 3rd Edition, 2008, McGraw-Hill

Paper VII/Subject Name: Principles of Management and Organizational Behavior		**Subject Code: BSA022C505**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand about the principles of management and their application to the functioning of an organization.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the managerial functions like planning, and have same basic knowledge on international aspect of management	**BT 2**
CO 2	**Build** the ability to direct, leadership and communicate effectively	**BT 3**
CO3	**Analyze** the behavior of individuals and groups in organizations in terms of the key factors that influence organizational behavior.	**BT 4**
CO 4	**Assess** the potential effects of organizational-level factors (such as structure, culture and change) on organizational behavior	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction To Management and Organizations**	Definition of Management – Science or Art, Manager vs. Entrepreneur, types of managers, managerial roles and skills. Evolution of Management-Scientific, human relations, system and contingency approaches, Types of Business organization, Sole proprietorship, partnership, company public and private sector enterprises, Organization culture and Environment, Current trends and issues in Management.	**09**
II	**Planning and Decision Making**	Nature and purpose of planning, planning process, types of planning, objectives, setting objectives, policies. Planning premises, Strategic Management, Planning Tools and	**09**

		Techniques, Decision making steps and process.	
III	**Organization and Human Resource Management**	Organizing- Nature and purpose, Formal and informal organization, organization chart, organization structure, types, Line and staff authority, departmentalization, delegation of authority, centralization and decentralization, Job Design. Human Resource Management- HR Planning, Recruitment, selection, Training and Development, Performance Management, Career planning and management.	**09**
IV	**Direction and Control**	Directing-Foundations of individual and group behavior, motivation, motivation theories, motivational techniques, job satisfaction, job enrichment. Leadership- types and theories of leadership, communication, process of communication, barrier in communication, effective Communication, Communication and IT. Controlling- System and process of controlling, budgetary and non-budgetary control techniques, use of computers and IT in Management control, Productivity problems and management, control and performance, direct and preventive control, reporting.	**09**
		TOTAL	**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Management,* Stephen P. Robbins and Mary Coulter, 13th Edition, 2017, Prentice Hall India Pvt. Ltd.
2. *Fundamentals of Management,* Stephen A. Robbins, David A. Decenzo and Mary Coulter, 9th Edition, 2016, Pearson Education India.

Reference Books:

1. Robert Kreitner and Mamata Mohapatra,*Management*, 1st Edition, 2008, Dreamtech Press.
2. Harold Koontz and Heinz Weihrich, *Essentials of Management: An International, Innovation and Leadership Perspective,* 10th Edition, 2015, Tata McGraw Hill.
3. Tripathy P. C. & Reddy P. N., *Principles of Management*, 4th Edition, 2010, Tata McGraw Hill.
4. J. P. Pathak, *Fundamentals of Management,* 1st Edition, 2014, Vikas Publishing House.
5. Robert N. Lussier, *Management Fundamentals Concepts, Applications, Skill Development,* 5th Edition, 2012, Cengage Publications.

Paper VI/Subject Name: Prompt Engineering		**Subject Code: CSE022G506**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Course Objective:

The objectives of the course are to equip students with the knowledge and practical skills required to design, optimize, and ethically apply prompt engineering techniques for Generative AI and Large Language Model based applications.

Prerequisites: None

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** the architecture and working principles of Generative AI and LLMs.	**BT 1 & 2**
CO 2	**Construct** effective prompts using zero-shot, few-shot, chain-of-thought, and role-based prompting methods.	**BT 3**
CO 3	**Analyse** and optimize prompts for accuracy, relevance, creativity, and safety.	**BT 4**
CO 4	**Evaluate** ethical, legal, and societal implications of AI-generated content and automation systems.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Foundations of Generative AI and Prompt Engineering**	Introduction to Artificial Intelligence and Generative AI, Evolution of Large Language Models (LLMs), Fundamentals of Prompt Engineering, Tokens, Context Windows, Temperature, Top-p Sampling, Transformer Architecture Basics, Working of GPT, Gemini, Claude, and Open-source LLMs, Types of Prompts: Instruction Prompts, Contextual Prompts, Role-based Prompts, Conversational Prompts, Prompt Design Principles, AI-assisted Learning and Productivity Applications	**15**
II	**Advanced Prompting Techniques**	Zero-shot Prompting, Few-shot Prompting, Chain-of-Thought Prompting, Self-consistency Prompting, ReAct Framework, Tree-of-Thought Prompting, Prompt Chaining Techniques, Structured Output Generation, Prompt Templates and Reusable Frameworks, Hallucination Detection and Mitigation, AI Output Validation Techniques	**15**
III	**Multimodal AI and AI-Assisted Development**	Introduction to Multimodal AI Systems, Text-to-Image Prompting, Image Understanding and Visual Prompting, Prompting for Audio and Video Generation, AI-assisted Coding using GitHub Copilot and LLMs, Code Generation and Debugging Prompts, Prompt Engineering for Data Analysis, AI Agents and Autonomous Workflows, Prompt APIs and Integration Basics, Domain-specific Prompting: Healthcare, Education, Agriculture, Business Analytics	**15**
IV	**Ethics, Security, Evaluation, and Future Trends**	Responsible and Ethical AI, Bias and Fairness in AI Models, Privacy, Security, and Prompt Injection Attacks, AI Governance and Regulatory Frameworks, Intellectual Property and Copyright Issues, Evaluating AI-generated Outputs, Benchmarking Prompt Performance, Human-in-the-loop Systems, Emerging Trends in Prompt Engineering, Future of AI-Augmented Computing	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Co-intelligence: Living and working with AI,* Ethian Mollick, E., 2024, Portfolio.
2. *Demystifying artificial intelligence for the enterprise,* Geerling, W., 2023, Routledge.

Reference Books:

1. White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, M., Spencer-Smith, J., & Schmidt, D. C., *A prompt pattern catalog to enhance prompt engineering with ChatGPT,* 2023, Vanderbilt University.
2. Marr, B., *Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society,* 2023, Wiley.

6.7 Detailed Syllabus of 6th Semester

Paper I/Subject Name: Computer Networks		**Subject Code: CSE022C601**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamental concepts and reference models of computer networks, apply error detection, flow control, and congestion control techniques in network protocols, etc.

Prerequisites: Concepts of Data Communication

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the layered architecture of computer networks and functions of each layer.	**BT 1 & 2**
CO 2	**Apply** subnetting, IP addressing, and routing algorithms in network design.	**BT 3**
CO 3	**Analyze** the operation of error control, flow control, and congestion control mechanisms.	**BT 4**
CO 4	**Evaluate** performance and reliability of network protocols and simulate them using tools.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Data Link Layer and Medium Access Sub-layer**	Design issues, Framing, Error detection and correction codes: checksum, CRC, hamming code, Data link protocols for noisy and noiseless channels, Sliding Window Protocols: Stop & Wait ARQ, Go-back-N ARQ, Selective repeat ARQ, Data link protocols: HDLC and PPP Static and dynamic channel allocation, Random Access: ALOHA, CSMA protocols, Controlled Access: Polling, Token Passing, IEEE 802.3 frame format, Ethernet cabling, Manchester encoding, collision detection in 802.3, Binary exponential back off algorithm	**09**
II	**Network Layer**	Design issues, IPv4 classful and classless addressing, subnetting, Routing algorithms: distance vector and link state routing, Congestion control: Principles of Congestion Control, Congestion prevention policies, Leaky bucket and token bucket algorithms	**09**
III	**Transport Layer**	Elements of transport protocols: addressing, connection establishment and release, flow control and buffering, multiplexing and de-multiplexing, crash recovery, introduction to TCP/UDP protocols and their comparison	**09**
IV	**Application Layer**	World Wide Web (WWW), Domain Name System (DNS), E-mail, File Transfer Protocol (FTP), SMTP, HTTP, Introduction to Network security	**09**
		TOTAL	**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data and Computer Communication*, William Stallings, 10th Edition, 2013, PHI.
2. *Data Communications and Networking*, Behrouz A Forouzan, 4th Edition, 2017, Tata McGraw Hill
3. *Computer Networks*, Tannenbaum, 3rd Edition, 1996, Pearson Education.

Reference Books:

1. L.L. Peterson & B.S. Davie, *Computer Networks: A Systems Approach*, 5th Edition, 2011, Morgan Kaufmann
2. Anuranjan Misra, *Computer Networks*, 2006, Acme Learning, Morgan Kaufman Publication, New Delhi

Paper II/Subject Name: Computer Networks Lab		**Subject Code: CSE022C611**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide hands-on experience in implementing and simulating network protocols and algorithms, enable students to design and analyze network topologies and configurations, develop the ability to write socket programs for client-server communication, etc.

Prerequisites: Concepts of Data Communication

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** understanding of protocol behavior and network performance through analysis and simulation	**BT 1 & 2**
CO 2	**Write** and execute socket programs for TCP and UDP communication	**BT 3**
CO 3	**Analyze** and simulate different network topologies using simulation tools	**BT 4**
CO 4	**Debug** test, and document network applications and simulations effectively	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. To study various topologies for establishing computer networks.
2. To learn the usage of various basic tools (crimping, krone etc.) used in establishing a LAN.
3. To familiarize with switch and hub used in networks.
4. To learn the usage of connecters and cables (cabling standards) used in networks
5. To make certain copper and fiber patch cords using different standards.
6. To familiarize with routers & bridges
7. Use commands like ping, ipconfig for trouble shooting network related problems.
8. NIC Installation & Configuration (Windows/Linux)
9. TCP/UDP Socket Programming
10. Multicast & Broadcast Sockets

11. Develop a program to compute the Hamming Distance between any two code words.
12. Develop a program to compute checksum for an _m' bit frame using a generator polynomial.
13. IPC (Message queue)
14. Implementation of a Prototype Multithreaded Server
    a. Implementation of o Data Link Layer Flow Control Mechanism (Stop & Wait, Sliding Window)
    b. Data Link Layer Error Detection Mechanism (Cyclic Redundancy Check)
    c. Data Link Layer Error Control Mechanism (Selective Repeat, Go Back N)

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data and Computer Communication*, William Stallings, 10th Edition, 2013, PHI.
2. *Data Communications and Networking*, Behrouz A Forouzan, 4th Edition, 2017, Tata McGraw Hill
3. *Computer Networks*, Tannenbaum, 3rd Edition, 1996, Pearson Education.

Reference Books:

1. L.L. Peterson & B.S. Davie, *Computer Networks: A Systems Approach*, 5th Edition, 2011, Morgan Kaufmann
2. Anuranjan Misra, *Computer Networks*, 2006, Acme Learning, Morgan Kaufman Publication, New Delhi

Paper III/Subject Name: Introduction to Deep Learning		**Subject Code: ARI022C602**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective

The objectives of the course are to provide the basic concept of neural networks and deep learning architectures, forward and backward propagation, loss functions, and optimization, evaluate CNNs, RNNs, and advanced deep learning models etc.

Prerequisites: Linear Algebra and Calculus, Probability and Statistics, Programming in Python, Basics of Machine Learning

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the architecture and learning process of neural networks.	**BT 1 & 2**
CO 2	**Apply** deep learning models using backpropagation and optimization techniques.	**BT 3**
CO 3	**Analyze** convolutional and recurrent neural networks.	**BT 4**
CO 4	**Assess** the performance of deep learning models on various datasets.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Introduction to deep learning and applications, Perceptron model and limitations, Feedforward Neural Networks (FNN), Activation functions: ReLU, Sigmoid, Tanh, Softmax, Loss functions: MSE, Cross-Entropy, Forward and Backward Propagation, Gradient Descent and variants (SGD, Adam, RMSprop)	**09**
II	**Deep Neural Networks and Regularization**	Deep Feedforward Networks and Vanishing Gradients, Batch normalization and Dropout, Weight initialization techniques, Hyperparameter tuning: learning rate, batch size, epochs, Model evaluation: accuracy, precision, recall, F1, confusion matrix, Overfitting and underfitting	**09**
III	**CNNs and RNNs**	Convolutional Neural Networks (CNN): Convolution, pooling, padding, Architecture: LeNet, AlexNet, VGG, ResNet, Applications in image classification and object detection Recurrent Neural Networks (RNN): Basics, vanishing gradients, LSTM and GRU, Applications in time-series and NLP	**09**
IV	**Advanced Topics**	Transfer learning and pretrained models (e.g., VGG, ResNet, BERT), Generative Adversarial Networks (GANs): Architecture, training challenges, Attention mechanism and Transformers (introductory level), Applications in computer vision, NLP, speech, healthcare, Model deployment using Flask/TensorFlow Lite, Ethical issues in deep learning (bias, fairness, explainability)	**09**
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Deep Learning,* Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2016, MIT Press
2. *Deep Learning: A Practitioner's Approach,* Josh Patterson, Adam Gibson, 1st Edition, 2017, O'Reilly

Reference Books:

1. Charu C. Aggarwal, *Neural Networks and Deep Learning*, 1st Edition, 2018, Springer

Paper IV/Subject Name: Introduction to Deep Learning Lab		**Subject Code: ARI022C612**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective

The objectives of the course are to provide hands-on experience in designing, implementing, and training deep neural networks, familiarize students with deep learning frameworks such as TensorFlow, Keras, or PyTorch, enable students to apply deep learning techniques to solve real-world problems in vision, text, or data, etc.

Prerequisites: Linear Algebra and Calculus, Probability and Statistics, Programming in Python, Basics of Machine Learning

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** understanding of practical issues like overfitting, regularization, and hyperparameter tuning	**BT 1 & 2**
CO 2	**Apply** Convolutional Neural Networks (CNNs) for image-related tasks	**BT 3**
CO 3	**Analyze** Recurrent Neural Networks (RNNs) or LSTM networks to sequence data	**BT 4**
CO 4	**Assess** the performance of deep learning models on various datasets.	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Introduction to NumPy, Pandas, and Matplotlib: Basics of Python for deep learning, Loading and preprocessing data
2. Implement a Perceptron and Feedforward Neural Network: Manual implementation using NumPy, XOR classification using MLP
3. Build an ANN using Keras or PyTorch, MNIST digit classification, Visualize loss and accuracy curves
4. Experiment with Different Activation and Loss Functions, Use ReLU, Sigmoid, Softmax, Compare MSE vs CrossEntropy
5. Optimization Techniques and Regularization, SGD, Adam optimizers, Dropout and Batch Normalization
6. Convolutional Neural Networks (CNNs), Build a CNN for CIFAR-10 or Fashion MNIST, Visualize feature maps
7. Transfer Learning with Pretrained Models, Use VGG16 or ResNet for classification, Fine-tune the final layers
8. Recurrent Neural Networks (RNNs): Character-level text generation, Basic RNN vs LSTM comparison
9. LSTM for Sentiment Analysis, IMDB or Twitter sentiment dataset, Embedding and sequence padding
10. Hyperparameter Tuning, Grid search or random search, Effect of learning rate and batch size
11. Evaluate Model Performance, Confusion matrix, precision, recall, F1-score, ROC-AUC curve

12. Mini Project: Students select a dataset and apply end-to-end deep learning, Example: plant disease detection, emotion recognition, etc.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Deep Learning,* Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2016, MIT Press
2. *Deep Learning: A Practitioner's Approach,* Josh Patterson, Adam Gibson, 1st Edition, 2017, O'Reilly

Reference Books:

1. Charu C. Aggarwal, *Neural Networks and Deep Learning*, 1st Edition, 2018, Springer

Paper V/Subject Name: Software Engineering		**Subject Code: CSE022C603**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain the fundaments of software engineering principles and practices, including project management, configurations management, requirements definition, system analysis, design, testing, and deployment.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **demonstrate** the fundamental concepts of software engineering, software process models, and development life cycles.	**BT 1 & 2**
CO 2	**Identify** software requirements and develop software specifications using functional and non-functional requirements.	**BT 3**
CO 3	**Analyze** risk factors, cost estimation, and scheduling using various estimation and modeling techniques.	**BT 4**
CO 4	**Evaluate** and select appropriate software testing strategies, maintenance models, and quality assurance standards.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Introduction to Process Models and**	Importance of Software Project Management, Activities Methodologies, Categorization of Software Projects, Setting	**09**

	Software Requirement Specification	objectives, Software life cycle models: Waterfall, prototyping, Evolutionary, Spiral models and Agile Model. Software Requirements: Functional and non-functional requirements, user requirements, system requirements, interface specification, the software requirements document. Requirements engineering process: Feasibility studies, requirements elicitation and analysis, requirements validation, requirements management.	
II	**Software Process Management, Activity Planning and Agile Development**	Project planning and control, Effort and Cost estimation techniques-LOC, Function Point, COCOMO, project scheduling using PERT and GANTT charts, Critical path (CRM) method, cost-time relations: Rayleigh-Norden results, Staffing Pattern,S oftware configuration management, Introduction to Agility- Agile methods – Extreme Programming – SCRUM – Managing interactive processes.	**09**
III	**Software Design and Risk Estimation**	Basics of Software Design, Procedural Design Methodology, Modularity, Cohesion, Coupling, DFD and Structure Chart, Object-Oriented concepts, Introduction to UML: Class and interaction Diagrams, Object-Oriented Analysis and Design, Object-oriented Software Modelling. Risk Management-Risk Identification, Risk Assessment, Risk Containment	**09**
IV	**Software Testing, Maintenance and Reuse**	Software testing fundamentals-Internal and external views of Testing-white box testing – basis path testing-control structure testing-black box testing-Regression Testing – Unit Testing – Integration Testing – Validation Testing – System Testing And Debugging –Software Implementation Techniques: Coding practices-Refactoring-Maintenance and Reengineering-BPR model-Reengineering process model-Reverse and Forward Engineering. Characteristics of Software Maintenance, Software Reverse Engineering, Software Maintenance Process Models, Estimation of maintenance cost, Software Reuse.	**09**
Total			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum/ Tutorial**	**Experiential Learning**
3* 12 NCH = 66 NCH	--	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Software Project Management,* Bob Hughes, Mike Cotterell and Rajib Mall, 5th Edition, 2012, Tata McGraw Hill, New Delhi

Reference Books:

1. Kieron Conway, *Software Project Management: From Concept to Deployment,* 1st Edition, 2000, Dreamtech Press.
2. S. A. Kelkar, *Software Project Management: A Concise Study,* 3rd Edition, 2012, PHI Publication.

Paper VI/Subject Name: Advanced Deep Learning (PEC-I)		**Subject Code: ARI022D607**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the concept advanced neural network architectures and deep generative models, attention, transformers, and large-scale pretraining, application of deep learning in specialized domains like vision, language, and multimodal systems, etc.

Prerequisites: AI/ML Fundamentals

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **demonstrate** generative models like GANs and VAEs.	**BT 1 & 2**
CO 2	**Utilize** advanced architectures like ResNet, DenseNet, and Inception.	**BT 3**
CO 3	**Analyze** and present recent advances and research trends in deep learning.	**BT 4**
CO 4	**Evaluate** and optimize models for interpretability, robustness, and deployment	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Advanced NN Architectures**	Deep CNN Architectures: ResNet, DenseNet, InceptionNet, MobileNet, EfficientNet for edge AI, Network engineering: skip connections, depthwise convolutions, residual learning, Neural architecture search (NAS) – basics, Attention Mechanisms (intro) in CNNs	**12**
II	**Generative Models**	Autoencoders and Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs): Vanilla GAN, DCGAN, CycleGAN, Training challenges: mode collapse, stability, Self-supervised learning basics, Contrastive learning and SimCLR	**12**
III	**Transformers**	Sequence modeling limitations of RNNs, Attention Mechanism and Self-Attention, Transformers: Encoder, Decoder, Positional Encoding, BERT, GPT, and their variants, Transfer learning and fine-tuning in NLP & vision (e.g., ViT), Vision-Language models: CLIP, BLIP (overview)	**12**

IV	**Deployment and Interpretability**	Model interpretability: SHAP, LIME, Grad-CAM, Adversarial attacks and defenses, Fairness, bias, and ethics in deep learning, Model compression: pruning, quantization, distillation, Real-time deployment using TensorFlow Lite, ONNX, Trends: diffusion models, foundation models, multimodal AI	**12**
TOTAL			**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Deep Learning,* Ian Goodfellow, Yoshua Bengio, Aaron Courville, 2016, MIT Press
2. *Deep Learning: A Practitioner's Approach,* Josh Patterson, Adam Gibson, 1st Edition, 2017, O'Reilly

Reference Books:

1. Charu C. Aggarwal, *Neural Networks and Deep Learning*, 1st Edition, 2018, Springer

Paper VII/Subject Name: Natural Language Processing (PEC-II)		**Subject Code: ARI022D608**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide the student with knowledge of various levels of analysis, language modelling involved in NLP.

Prerequisites: Concepts of Automata Theory

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** the fundamental concepts of Natural Language Processing.	**BT 2**
CO 2	**Solve** the NLP tasks using various categories of algorithms.	**BT 3**
CO 3	**Analyze** the algorithms applied	**BT 4**
CO 4	**Evaluate** the algorithms applied	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**

I	**Overview and Language Modeling**	Overview: Origins and challenges of NLP-Language and Grammar-Processing Indian Languages-NLP Applications-Information Retrieval. Language Modeling: Introduction-Various Grammar-based Language Models-Statistical Language Model	**12**
II	**Word Level, Syntactic and Semantic Analysis**	Word Level Analysis: Introduction- Regular Expressions-Finite-State Automata-Morphological Parsing-Spelling Error Detection and correction-Words and Word classes-Part-of Speech Tagging. Syntactic Analysis: Introduction-Context-free Grammar-Constituency Parsing-Probabilistic Parsing Semantic Analysis: Introduction- Meaning Representation-Lexical Semantics Ambiguity-Word Sense Disambiguation. Discourse Processing: Introduction- cohesion-Reference Resolution Discourse Coherence and Structure	**12**
III	**Natural Language Generation and Machine Translation**	Natural Language Generation: Introduction-Architecture of NLG Systems Generation Tasks and Representations-Application of NLG. Machine Translation: Introduction-Problems in Machine Translation Characteristics of Indian Languages- Machine Translation Approaches-Translation involving Indian Languages	**12**
IV	**Information Retrieval and Lexical Resources**	Information Retrieval: Introduction-Design features of Information Retrieval Systems-Classical, Non-classical, Alternative Models of Information Retrieval - Evaluation Lexical Resources: Introduction-WordNet-FrameNet-Stemmers-POS Tagger Research Corpora	**12**
TOTAL			**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Natural Language Processing and Information Retrieval*, Tanveer Siddiqui, U.S. Tiwary, 1st Edition, 2008, Oxford University Press

Reference Books:

1. Daniel Jurafsky and James H Martin, *Speech and Language Processing: An introduction to Natural Language Processing, Computational Linguistics and Speech Recognition*, 2nd Edition, 2008, Prentice Hall.
2. James Allen, Bejamin Cummings, *Natural Language Understanding*, 2nd Edition, 1995, Pearson.

L-T-P-C – 3-0-0-3 **Credit Units: 03** **Scheme of Evaluation: T**

Course Objective:

The objectives of the course are to develop entrepreneurial mindset, innovation capabilities, and startup management skills among students. The course aims to familiarize learners with the startup ecosystem, business model development, funding mechanisms, intellectual property rights, and technology commercialization

Prerequisites: None

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** entrepreneurship, innovation, and startup ecosystems.	**BT 1 & 2**
CO 2	**Identify** business opportunities and develop sustainable business models.	**BT 3**
CO 3	**Analyze** startup funding, legal, and intellectual property aspects.	**BT 4**
CO 4	**Develop** and present a startup business plan for technology-driven ventures.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Entrepreneurship and Startup Ecosystem**	Concept, characteristics, and importance of entrepreneurship, Entrepreneurial mindset and competencies, Types of entrepreneurships: Technology, Social, Rural, and Digital Entrepreneurship, Innovation and Design Thinking fundamentals, Startup ecosystem in India, Government initiatives: Startup India, Make in India, Digital India, Atal Innovation Mission, Success and failure stories of startups	**12**
II	**Opportunity Identification and Business Model Development**	Creativity and idea generation techniques, Problem identification and opportunity recognition, Market research and customer discovery, Value proposition design, Business Model Canvas (BMC) Lean Startup methodology, Minimum Viable Product (MVP), Customer validation and product-market fit	**12**
III	**Startup Formation, Funding, and Intellectual Property**	Steps in starting a venture, Forms of business organization, Startup registration and legal compliance, Intellectual Property Rights (IPR): Patents, Copyrights, Trademarks, Trade Secrets, Technology transfer and commercialization, Sources of startup funding: Bootstrapping, Angel Investors, Venture Capital, Crowdfunding, Government grants and incubators, Financial planning and revenue models	**11**

IV	**Startup Growth, Scaling, and Business Planning**	Marketing strategies for startups, Digital marketing and social media branding, Team building and leadership in startups, Risk management and business ethics, Scaling strategies for technology startups, Startup valuation basics, Pitch deck preparation, Business plan development and investor pitching	**10**
Total			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Entrepreneurship*, Robert D. Hisrich, Michael P. Peters and Dean A. Shepherd, 9th Edition, 2012, McGraw-Hill.
2. *Zero to One: Notes on Start Ups, or How to Build the Future,* Peter Thiel, 1st Edition, 2014, Crown Business.

Reference Books:

1. Eric Ries, *The Lean Startup*, 1st Edition, 2011, Crown Publishing.

6.8 Detailed Syllabus of 7th Semester

Paper I/Subject Name: Data and Visual Analytics in AI		**Subject Code: ARI022C701**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students learn data analytics concepts and visualization techniques, integrate AI/ML methods for extracting insights from structured and unstructured data, learn effective data storytelling and dashboarding for decision-making, etc.

Prerequisites: Basics of Python programming, Data Structures and Algorithms, Probability, Statistics, and Linear Algebra, Fundamentals of Machine Learning

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the role of data analytics and visualization in AI-based systems.	**BT 1 & 2**
CO 2	**Apply** preprocessing, transformation, and dimensionality reduction techniques on real-world data.	**BT 3**
CO 3	**Analyze** AI models and visualize their results.	**BT 4**
CO 4	**Evaluate** and communicate insights using interactive dashboards and visual tools.	**BT 5**

Detailed Syllabus

Modules	**Topics**	**Course Contents**	**Hours**
I	**Introduction**	Introduction to data analytics in AI, Data types, scales, and distributions, Data wrangling and preprocessing (missing data, outliers, normalization), Exploratory Data Analysis (EDA): histograms, boxplots, scatter plots, Introduction to data visualization principles (Tufte's theory, Gestalt principles), Tools: Python (Matplotlib, Seaborn, Plotly), Tableau/Power BI (overview)	**12**
II	**Dimensionality Reduction and Feature Engineering**	Correlation and Covariance matrices, Principal Component Analysis (PCA) and t-SNE, Linear Discriminant Analysis (LDA), Feature selection vs feature extraction, Feature importance visualization in ML models (e.g., SHAP, permutation importance), Case study: reducing dimensionality in image or text datasets	**12**
III	**Machine Learning and Visual Analytics**	Supervised learning recap: regression and classification, Unsupervised learning recap: clustering and association rules, Visualizing ML performance: ROC, confusion matrix, lift charts, Interactive visualizations for ML insights (using Dash, Streamlit), AI explanations: SHAP values, partial dependence plots, decision plots, Real-world AI use cases: fraud detection, healthcare analytics, customer segmentation	**12**
IV	**Storytelling, Dashboards and Ethics**	Data storytelling concepts: audience, message framing, design, Creating interactive dashboards using Tableau, Power BI, or Dash, Real-time data visualization (live APIs, IoT, log data), Visualizing NLP and time-series data, Ethical visualization: avoiding misrepresentation, bias, overfitting, Capstone mini-project: end-to-end analytics pipeline with visuals	**12**
TOTAL			**48**

Data and Visual Analytics in AI Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 10 Laboratory experiments based on the following-

- Installation and familiarization with Anaconda, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn and Plotly.
- Import datasets (CSV/Excel/JSON), inspect data types, summarize statistics, and perform data cleaning.
- Handle missing values, duplicate records, outliers, normalization and standardization of datasets.
- Perform Exploratory Data Analysis (EDA) using histograms, bar charts, boxplots, scatter plots, pair plots and heatmaps. 2
- Correlation analysis and covariance matrix visualization using Seaborn.
- Apply Principal Component Analysis (PCA) for dimensionality reduction and visualize transformed features.
- Perform feature selection using correlation analysis and feature importance techniques.
- Train basic machine learning models (Linear Regression, Logistic Regression, Decision Tree or Random Forest) and visualize model performance using confusion matrix and ROC curve.
- Generate model explanation visualizations using SHAP or permutation feature importance.
- Create interactive visualizations using Plotly and Dash (or Streamlit).
- Design an interactive dashboard using Tableau or Microsoft Power BI for a real-world dataset.
- Mini Project: Develop an end-to-end data analytics pipeline including preprocessing, visualization, machine learning, and dashboard/report presentation.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	2 * 15 = 30 NCH	45 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Visualizing Data,* Ben Fry, 1st Edition, 2016, O'Reilly
2. *Data Points: Visualization That Means Something,* Nathan Yau, Illustrated Edition, 2013, Wiley
3. *Data Science from Scratch,* Joel Grus, 1st Edition, 2015, O'Reilly

Reference Books:

1. Kaufman & Rousseeuw, *Finding Groups in Data: An Introduction to Cluster Analysis*, 1st Edition, 2005, Wiley
2. Anil Maheshwari, *Data Analytics Made Accessible*, Kindle Edition, 2025, Amazon/KDP

Paper II/Subject Name: Web Technology		**Subject Code: CSE022C702**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide knowledge on the basic web concepts, scripting languages and Internet protocols.

Prerequisites: Concepts of Databases and Object-Oriented Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **understand** the basic concept of web development	**BT 1 & 2**
CO 2	**Apply** the concepts learnt to develop simple web applications	**BT 3**
CO 3	**Assess** two web applications based on various design factors.	**BT 4**
CO 4	**Evaluate** the working of various web applications.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction, To Web Technology**	World Wide Web: Introduction to TCP/IP and WAP, DNS, Email, TelNet, HTTP and FTP. Introduction to Browser and search engines, Working of the search engines, Miscellaneous Web Browser details, Introduction to Web Servers: Features of web servers, caching, case study-IIS, Apache, Configuring web servers.Internet Principles – Basic Web Concepts – Client/Server model – retrieving data from Internet – HTM and Scripting Languages – Standard Generalized Mark –up languages – Next Generation – Internet –Protocols and Applications.	**09**
II	**HTML,CSS, Java Script**	Web Pages - types and issues, tiers; comparisons of Microsoft and java technologies, WWW-Basic concepts, web client and web server, http protocol (frame format), universal resource locator (URL), HTML different Tags, sections, image & pictures, listings, tables, frame, frameset, form. The need of dynamic web pages; an overview of DHTML, cascading style sheet (CSS), comparative studies of different technologies of dynamic page creation.Java Script : Data types, variables, operators, conditional statements, array object, date object, string object, Dynamic Positioning and front end validation, creating rollovers, building smarter forms, Event Handling, working with cookies, DOM, node and objects, creating sliding menu, pop-up menu, slideshow with caption	**09**
III	**XML and AJAX**	XML – Server side includes – communication – DTD – Vocabularies – DOMmethods – Introduction of XML, Validation of XML documents, DTD, Ways to use XML, XML for data files, HTML Vs XML, Embedding XML into HTML documents, Converting XML to HTML for Display, Rewriting HTML as XML, Firewalls– Proxy Servers.AJAX technologies, Action, XML HttpRequest database operations, security, issues	**09**
IV	**J2SE, J2EE, Servlet and JSP**	Data Types, Arrays, Type Casting, Classes and Objects, Inheritance, Interfaces, Exception Handling, Multithreading,J2EE as a framework, Client Server Traditional model, Comparison amongst 2-tier, 3-tier and N-tier Architectures, Thin and Thick Clients. J2EE Servlet 2.x Specification, Writing small Servlet Programs, Deployment Descriptor, Inter Servlet Collaboration, Session:	**09**

		Definition, State on web, Different ways to track sessions.JSP Technology Introduction-JSP and Servlets- Running JSP Applications Basic JSP- JavaBeans Classes - Support for the Model-View- Controller Paradigm- Case Study- Related Technologies.	
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Internet and World Wide Web How to program*,Deitel H.M. and Deitel P.J, 4th Edition, 2012, Pearson International, New Delhi.
2. *Web Technology,* Gopalan N.P. and Akilandeswari J., 2nd Edition, 2014. Prentice Hall of India.
3. *Java How to Program*, Paul Dietel and Harvey Deitel, 8th Edition, 2014, Prentice Hall of India.

Reference Books:

1. Uttam K. Roy, *Web Technologies*, 2011, Oxford University Press.
2. Godbole A. S. & Kahate A., TMH, *Web Technologies*, 2ndEdition, 2006, TMH

Paper III/Subject Name: Web Technology Lab		**Subject Code: ARI022C712**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide knowledge on the basic web concepts, scripting languages and Internet protocols.

Prerequisites: Concepts of Databases and Object-Oriented Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **understand** the basic concept of web development	**BT 1 & 2**
CO 2	**Apply** the concepts learnt to develop simple web applications	**BT 3**
CO 3	**Assess** two web applications based on various design factors.	**BT 4**
CO 4	**Evaluate** the working of various web applications.	**BT 5**

Detailed Syllabus:

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Web page design: Designing web pages with HTML- use of tags, hyperlinks, URLs, tables, text formatting, graphics & multimedia, imagemap, frames and forms in web pages.
2. Use of Cascading Style Sheet in web pages.
3. Creating interactive and dynamic web pages with JavaScript: JavaScript overview; constants, variables, operators, expressions & statements; user-defined & built-in functions; client-side form validation; using properties and methods of built-in objects.
4. Extensible Markup Language (XML): Introduction- using user-defined tags in web pages; displaying XML contents; XML DTDs; use of XSL.
5. Server-side scripting: overview of CGI, ASP, and JSP.
6. Server side scripting using PHP; PHP basics, HTML form data handling, Web database connectivity- introduction to ODBC; PHP with database connectivity.
   Exposure to Advanced Web Technologies (as far as possible; not to be made compulsory): Distributed Object based models- DCOM, CORBA, EJB; Web services and Related Technologies- ISAPI, SOAP, UDDI, WSDL; Other Advanced Web Technologies- AJAX, ISAPI, .NET. Web Security

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Internet and World Wide Web How to program*,Deitel H.M. and Deitel P.J, 4th Edition, 2012, Pearson International, New Delhi.
2. *Web Technology,* Gopalan N.P. and Akilandeswari J., 2nd Edition, 2014. Prentice Hall of India.
3. *Java How to Program*, Paul Dietel and Harvey Deitel, 8th Edition, 2014, Prentice Hall of India.

Reference Books:

1. Uttam K. Roy, *Web Technologies*, 2011, Oxford University Press.
2. Godbole A. S. & Kahate A., TMH, *Web Technologies*, 2ndEdition, 2006, TMH

Paper IV/Subject Name: Computer Vision (PEC-III)		**Subject Code: ARI022D701**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamentals of digital image processing and vision systems, explore image transformation, filtering, edge detection, and feature extraction techniques, implement machine learning and deep learning approaches for object recognition, etc.

Prerequisites: Linear Algebra, Probability & Statistics, Python Programming, Basics of Machine Learning and Deep Learning

Course Outcomes

On successful completion of the course the students will be able to:

SI No	Course Outcome	Blooms Taxonomy Level
CO 1	**Describe** and **explain** core concepts of image formation, transformation, and processing.	**BT 1 & 2**
CO 2	**Apply** techniques for image filtering, edge detection, and feature extraction.	**BT 3**
CO 3	**Analyze** image classification and object detection pipelines.	**BT 4**
CO 4	**Evaluate** and compare vision models using performance metrics.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Image Processing Fundamentals**	Introduction to computer vision and its applications, Image formation, pixel operations, Color spaces: RGB, HSV, Grayscale, Image thresholding, histogram equalization, Smoothing and filtering: Gaussian, median, bilateral, Edge detection: Sobel, Canny, Laplacian, Geometric transformations: translation, scaling, rotation, affine	**12**
II	**Feature Detection and Matching**	Interest point detection: Harris, FAST, Local descriptors: SIFT, SURF, ORB, Feature matching using distance metrics, Homography and image stitching, Motion estimation, optical flow basics, Object tracking: Kalman filter, Mean shift, Camshift	**12**
III	**ML and DL in Vision**	Classical classification: KNN, SVM for image data, Introduction to CNNs: architecture, layers, filters, Popular CNN models: LeNet, AlexNet, VGG, ResNet, Object detection models: R-CNN, Fast R-CNN, YOLO, SSD, Image segmentation: semantic and instance segmentation (Mask R-CNN)	**12**
IV	**Application, Ethics and Deployment**	Applications: facial recognition, OCR, autonomous driving, healthcare, Real-time video analysis and gesture recognition, Vision in robotics and AR/VR, Explainable AI in computer vision (Grad-CAM), Ethical issues: bias, surveillance, privacy, Deploying models using OpenCV, TensorFlow Lite, ONNX	**12**
TOTAL			**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Computer Vision: Algorithms and Application,* Richard Szeliski, 1st Edition, 2010, Springer

Reference Books:

1. Dana H. Ballard and Christopher M. Brown, *Computer Vision*, 1st Edition, 1982, Prentice Hall.
2. Theo Pavlidis, *Algorithms for Graphics and Image Processing*, 1st Edition, 1982, Springer-Verlag Berlin Heidelberg

Paper V/Subject Name: Reinforcement Learning (PEC-IV)		**Subject Code: ARI022D702**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to introduce studentsto the fundamental concepts and mathematical formulation of reinforcement learning, explore dynamic programming, Monte Carlo, and temporal-difference learning methods, implement RL algorithms and apply them to games and control problems, etc.

Prerequisites: Linear Algebra and Probability, Python Programming, Basics of Machine Learning & Neural Networks (for advanced topics)

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **understand** the core concepts of agents, environments, and rewards in reinforcement learning.	**BT 1 & 2**
CO 2	**Apply** value-based and policy-based methods to sequential decision-making problems.	**BT 3**
CO 3	**Analyze** RL algorithms using Bellman equations and convergence properties.	**BT 4**
CO 4	**Evaluate** and implement deep RL algorithms for complex environments.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Introduction**	Agent-Environment Interaction, Markov Decision Processes (MDP), Rewards, Policies, Returns, Value Functions (State-value, Action-value), Bellman Expectation Equations, Exploration vs Exploitation, Gridworld examples	**12**
II	**Dynamic Programming & TD Learning**	Policy Evaluation and Iteration, Value Iteration, Monte Carlo Methods: First-visit, Every-visit, Temporal-Difference (TD) Learning, SARSA and Q-Learning, ε-Greedy and Softmax policies	**12**
III	**Policy Gradient and Actor-Critic Methods**	Policy Gradient Theorem, REINFORCE Algorithm, Actor-Critic Architecture, Advantage Estimation, Trust Region Policy Optimization (TRPO), Proximal Policy Optimization (PPO), Applications in control and robotics	**12**

IV	**Deep Reinforcement Learning & Applications**	Deep Q-Networks (DQN), Target networks and experience replay, Double DQN, Dueling DQN, Deep Deterministic Policy Gradient (DDPG), Multi-Agent RL (basics), Applications in games (Atari, Chess, Go), robotics, recommendation systems, Ethics and safety in RL	**12**
TOTAL			**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Reinforcement Learning: An Introduction,* Richard S. Sutton and Andrew G. Barto, 2nd Edition, 2018, MIT Press
2. *Reinforcement Learning: Industrial Applications of Intelligent Agents,* Phil Winder, 2020, O'Reilly

Reference Books:

1. Laura Graesser & Wah Loon Keng, *Foundations of Deep Reinforcement Learning*, 1st Edition, 2020, Pearson
2. Alexander L. Strehl, *Reinforcement Learning: Theory and Algorithms*, 2011, Springer

6.9 Detailed Syllabus of 8th Semester

Paper I/Subject Name: Predictive Modelling & Optimization Techniques		**Subject Code: ARI022C801**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students learn various predictive modelling techniques and their real-world applications, equip students with optimization techniques for enhancing the performance of predictive models, impart practical skills in building, evaluating, and tuning models using tools like Python, R, or MATLAB, etc.

Prerequisites: Probability and Statistics, Linear Algebra and Calculus, Programming with Python/R, Basic understanding of Machine Learning

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **interpret** fundamental predictive modelling concepts and statistical learning techniques.	**BT 1 & 2**
CO 2	**Apply** regression, classification, and ensemble methods to predictive tasks.	**BT 3**
CO 3	**Analyze** and compare model performance using relevant evaluation metrics.	**BT 4**
CO 4	**Evaluate** optimization techniques to improve model accuracy and efficiency.	**BT 5**

Detailed Syllabus

Modules	Topics	Course Contents	Hours
I	**Introduction**	Overview of predictive analytics and its applications, Statistical learning and data preparation, Types of predictive models: regression, classification, time-series, Linear and logistic regression, Bias-variance trade-off, Model validation: hold-out, cross-validation	**09**
II	**Advanced Predictive Modelling Techniques**	Decision Trees, Random Forests, Gradient Boosting (XGBoost, LightGBM), k-NN, SVM, Naïve Bayes, Model evaluation: Accuracy, Precision, Recall, F1, ROC-AUC, Model interpretability (LIME, SHAP), Handling imbalanced data and missing values, Feature engineering and selection techniques	**09**
III	**Optimization Techniques**	Introduction to optimization: objective function, constraints, Gradient Descent and its variants (SGD, Adam, RMSprop), Linear Programming and Integer Programming, Metaheuristic optimization: Genetic Algorithms, Particle Swarm Optimization, Hyperparameter tuning: Grid Search, Random Search, Bayesian Optimization	**09**
IV	**Applications and Case Studies**	Predictive analytics in finance, healthcare, marketing, and supply chain, Time series forecasting using ARIMA, Prophet, LSTM, AutoML tools and pipelines (e.g., TPOT, H2O AutoML, Scikit-learn pipelines), Deploying predictive models using Flask/Streamlit, Capstone project: End-to-end predictive modelling with optimization	**09**
TOTAL			**36**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 12 NCH = 36 NCH	--	54 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *The Elements of Statistical Learning,* Trevor Hastie, Robert Tibshirani, Jerome Friedman, 9th Edition, 2017, Springer
2. *Applied Predictive Modeling,* Max Kuhn & Kjell Johnson, 1st Edition, 2018, Springer

Reference Books:

1. S. Rajasekaran, *Optimization: Theory and Practice*, 4th Edition, 2009, PHI Learning
2. Amit Kumar Tyagi, *Data Analytics and Predictive Modeling*, 1st Edition, 2024, Wiley

Paper II/Subject Name: Predictive Modelling & Optimization Techniques Lab		**Subject Code: ARI022C811**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students learn various predictive modelling techniques and their real-world applications, equip students with optimization techniques for enhancing the performance of predictive models, impart practical skills in building, evaluating, and tuning models using tools like Python, R, or MATLAB, etc.

Prerequisites: Probability and Statistics, Linear Algebra and Calculus, Programming with Python/R, Basic understanding of Machine Learning

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **interpret** fundamental predictive modelling concepts and statistical learning techniques.	**BT 1 & 2**
CO 2	**Apply** regression, classification, and ensemble methods to predictive tasks.	**BT 3**
CO 3	**Analyze** and compare model performance using relevant evaluation metrics.	**BT 4**
CO 4	**Evaluate** optimization techniques to improve model accuracy and efficiency.	**BT 5**

Detailed Syllabus

Total Lab Hours for the semester = 24 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Data Exploration and Preprocessing: Load and clean a real-world dataset, handle missing, values, outliers, and encode categorical variables, Normalize/standardize features
2. Linear and Logistic Regression Models: Apply linear regression for prediction (e.g., house prices), Apply logistic regression on a binary classification problem, plot decision boundaries and residual errors
3. Decision Tree and Random Forest Classifiers: Build and visualize decision trees, Evaluate Random Forest using accuracy, F1-score, and feature importance, Compare overfitting/underfitting using training vs testing performance
4. Gradient Boosting with XGBoost and LightGBM: Implement XGBoost and LightGBM, Visualize learning curves and confusion matrix, Analyze feature importance and SHAP values

5. K-Nearest Neighbors and SVM Models: Apply k-NN and SVM to multi-class classification problems, Tune hyperparameters (K, kernel type), Visualize the effect of different kernels on decision boundaries
6. Hyperparameter Tuning using Grid Search and Random Search: Use GridSearchCV and RandomizedSearchCV, Compare performance metrics with default vs tuned models,Plot tuning curves
7. Metaheuristic Optimization – Genetic Algorithms: use DEAP/PyGAD or custom implementation, Optimize hyperparameters of a model using GA, Compare with Grid Search,
8. Linear and Integer Programming with PuLP/Excel Solver: Formulate and solve LP/IP problems for resource allocation, Use Python's PuLP or Excel Solver to find optimal solutions, Apply LP in model selection or feature reduction scenarios
9. Time Series Forecasting: Use ARIMA/Prophet/LSTM on stock or weather data, Plot predictions vs actual, Evaluate using MAE, RMSE
10. Model Explainability: Use SHAP, LIME for explaining Random Forest/XGBoost models, Visualize and interpret SHAP summary plots, Generate insights for stakeholders
11. Deployment of Predictive Model: Build a Streamlit/Flask app for a predictive task, Allow user input and display predictions and charts, Export and import trained model using joblib/pickle
12. Mini Project: Implement end-to-end predictive pipeline, Sample projects: churn prediction, energy forecasting, fraud detection, Documentation and demonstration

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
--	1 * 12 NCH = 12 NCH	18 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *The Elements of Statistical Learning,* Trevor Hastie, Robert Tibshirani, Jerome Friedman, 9th Edition, 2017, Springer
2. *Applied Predictive Modeling,* Max Kuhn & Kjell Johnson, 1st Edition, 2018, Springer

Reference Books:

1. S. Rajasekaran, *Optimization: Theory and Practice*, 4th Edition, 2009, PHI Learning
2. Amit Kumar Tyagi, *Data Analytics and Predictive Modeling*, 1st Edition, 2024, Wiley

Paper III/Subject Name: Cryptography and Network Security		**Subject Code: CSE022C802**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain the basics of cryptography, kinds of security threats in networks and to learn to find the vulnerabilities in programs and to overcome them and to teach about the models and standards for security.

Prerequisites: Concepts of Number Theory and Networking

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** and **illustrate** basic cryptographic algorithms, message and web authentication and security issues.	**BT 2**
CO 2	**Demonstrate** the current legal and ethical issues towards information.	**BT 2**
CO 3	**Identify** the applications of different protocol like SSL, TLS etc.	**BT 3**
CO 4	**Analyze** and **assess** the security services and mechanisms	**BT 4**

Detailed Syllabus

Modules	Topics	Course Contents	Hours
I	**Introduction**	Need for Security, Security Approaches, Principles of Security, Types of Attacks, Brute Force Attack, Encryption, Decryption, Crptosystem, Cryptographic Techniques: Substitution Ciphers, Transposition Ciphers, Product Ciphers, Stegenography, Block Cipher, Stream Cipher.	**12**
II	**Symmetric and Asymmetric Key Cryptography**	Overview, Algorithm Modes and Types, Data Encryption Standard: Simplified DES, The Strength of DES, Differential and Linear Cryptanalysis. Triple DES, Blowfish. Confidentiality using Conventional Encryption: Placement of Encryption Function, Traffic Confidentiality, Key Distribution, Random Number Generation. Modular Arithmetic, Public Key Cryptography and RSA: Principles of Public Key Cryptosystems, Difference with Symmetric Key Cryptography, The RSA Algorithms, Key Management, Diffie Hellman Key Exchange.	**12**
III	**Authentication Protocols**	Message Authentication: Authentication Requirements, Authentication Functions, Message Authentication Codes, MD5 Message Digest Algorithms, Digital Signatures and Authentication Protocols: Digital Signatures, Authentication Protocols, Digital Signature Standards.	**12**
IV	**Security Protocols**	Security Applications and Protocols- Authentication Applications: Secure HTTP, HTTPS, ERT, SSH, Kerberos. Email Security: PGP, S/MIME. IP Security: Overview, IPSec architecture.	**12**
TOTAL			**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Cryptography and Network Security*, Atul Kahate, 2nd Edition. 2003, Tata McGraw Hill.
2. *Cryptography and Network security,* Fourozan, 3rd Edition, 2007, McGraw Hill

Reference Books:

1. William Stallings, *Cryptography and Network Security*: *Principles and Practices*", 5th Edition, 2010, Prentice Hall.
2. Michael Howard, David LeBlanc, John Viega, *24 Deadly Sins of Software Security: Programming Flaws and How to Fix Them*, 1st Edition, 2009, Mc Graw Hill Osborne Media.

Paper IV/Subject Name: AI for Robotics (PEC-V)		**Subject Code: ARI022D801**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the integration of AI techniques in robotic systems, path planning, localization, and mapping algorithms, apply machine learning and computer vision for robotic perception and control, etc.

Prerequisites: Basic Programming (Python/C++), Linear Algebra, Probability, and Calculus, Fundamentals of AI / Machine Learning

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **understand** the foundations of robotic systems integrated with AI algorithms.	**BT 1 & 2**
CO 2	**Apply** probabilistic reasoning and search techniques to robot planning and navigation.	**BT 3**
CO 3	**Analyze** perception, localization, and mapping algorithms.	**BT 4**
CO 4	**Assess** machine learning and computer vision-based models for robotic perception and control.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Overview of robotics: components (sensors, actuators, controllers), Role of AI in robotics: perception, cognition, control, Agent-based models and environments, Search algorithms: BFS, DFS, A*, D*, Configuration space and motion planning, Robot Operating System (ROS) introduction	**12**
II	**Localisation, Mapping an Navigation**	Probabilistic robotics basics, Markov localization and particle filters, Kalman and Extended Kalman Filters (EKF), Simultaneous Localization and Mapping (SLAM), Path planning: RRT, PRM, Obstacle avoidance: potential fields, dynamic window approach	**12**

III	**Perception and Computer Vision**	Camera models and calibration, Feature detection and matching (SIFT, ORB), Depth estimation and stereo vision, Visual odometry and 3D reconstruction, Object detection: YOLO, SSD, Sensor fusion (LIDAR + vision)	**12**
IV	**Learning & Autonomous Decision Making**	Supervised and Reinforcement Learning for robotics, Q-Learning, SARSA, Deep Q Networks (DQN), Policy gradient and actor-critic methods, Behavior trees and task-level planning, Human-robot interaction (HRI), Ethics and safety in AI-driven robots	**12**
Total			**48**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 12 NCH = 48 NCH	--	72 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Probabilistic Robotics,* Sebastian Thrun, Wolfram Burgard, Dieter Fox, 2005, MIT Press
2. *Robotics, Vision and Control*, Peter Corke, 2nd Edition, 2017, Springer

Reference Books:

1. M. J. Mataric, *The Robotics Primer*, 2007, MIT Press
2. Sudeep Das, *AI and Machine Learning for Robotics*, BPB Publications
3. Francis X. Govers, *Artificial Intelligence for Robotics: Build intelligent robots that perform human tasks using AI techniques,* 2018, Packt Publishers